

O-RAN White Box Hardware Working Group Hardware Reference Design Specification for Outdoor Micro Cell with Split Architecture Option 7.2

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Revision History

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06/22/2022	v03.00	WG7	Expanded chapter 2.3.7 to include detailed information on thermal design. Added chapter 2.3.2.3 to provide details of optional internal interfaces within an O-RU. Updated header and footers.
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Chapter 1 Introductory Material

1.1 Scope

This Technical Specification has been produced by the O-RAN.org.

The contents of the present document are subject to continuing work within O-RAN WG7 and may change following formal O-RAN approval. Should the O-RAN.org modify the contents of the present document, it will be re-released by O-RAN Alliance with an identifying change of release date and an increase in version number as follows:

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y the second digit is incremented when editorial only changes have been incorporated in the document.

z the third digit included only in working versions of the document indicating incremental changes during the editing process. This variable is for internal WG7 use only.

The present document specifies system requirements and high-level architecture for the Outdoor Micro Cell for FR1 deployment scenario as specified in the Deployment Scenarios and Base Station Classes document [1].

In the main body of this specification (in any “chapter”) the information contained therein is informative, unless explicitly described as normative. Information contained in an “Annex” to this specification is always informative unless otherwise marked as normative.

1.2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

[1] ORAN-WG7.DSC.0-V02.00 Technical Specification, ‘Deployment Scenarios and Base Station Classes for White Box Hardware’. <https://www.o-ran.org/specifications>

[2] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[3] 3GPP TR 38.104: "NR; Base Station (BS) radio transmission and reception".
http://www.3gpp.org/ftp/Specs/archive/38_series/38.104/38104-g10.zip

[4] ORAN-WG4.CUS.0-v05.00 Technical Specification, ‘O-RAN Fronthaul Working Group Control, User and Synchronization Plane Specification’. <https://www.o-ran.org/specifications>

- [5] 3GPP TS 38.113: "NR: Base Station (BS) Electromagnetic Compatibility (EMC)".
http://www.3gpp.org/ftp/Specs/archive/38_series/38.113/38113-f80.zip
- [6] ORAN-WG7.OMC. HAR-v01.00 Technical Specification, 'Outdoor Micro Cell Hardware Architecture and Requirement (FR1) Specification'. <https://www.o-ran.org/specifications>.
- [7] "Data Plane Development Kit", <https://www.dpdn.org/>
- [8] "Baseband Device Library", https://doc.dpdn.org/guides/prog_guide/bbdev.html.
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- [20] Universal Chiplet Interconnect Express web site: <https://www.uciexpress.org/>

1.3 Definitions and Abbreviations

1.3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [2] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [2]. For the base station classes of Pico, Micro and Macro, the definitions are given in 3GPP TR 38.104 [3].

Integrated architecture: In the integrated architecture, the O-RU and O-DU are implemented on one platform. Each O-RU and RF front end is associated with one O-DU. They are then aggregated to O-CU and connected by F1 interface.

Split architecture: The O-RU and O-DU are physically separated from one another in this architecture. A switch may aggregate multiple O-RUs to one O-DU. O-DU, switch and O-RUs are connected by the fronthaul interface as defined in WG4.

1.3.2 Abbreviations

For the purposes of the present document, the abbreviations given in [2] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, as in [2].

3GPP	Third Generation Partnership Project
5G	Fifth-Generation Mobile Communications
7-2	Fronthaul interface split option as defined by O-RAN WG4, also referred to as 7-2x
ACLR	Adjacent Channel Leakage Ratio
ADC	Analog to Digital Converter
AGC	Automatic Gain Control
AIB	Advanced Interface Bus
AISG	Antenna Interface Standards Group
ASIC	Application Specific Integrated Circuit
BBDEV	Baseband Device
BH	Backhaul
BIOS	Basic Input/Output System
BMC	Baseboard Management Controller
BoW	Bunch of Wires

1	BS	Base Station
2	BW	Bandwidth
3	C2C	Chip to Chip Interface
4	CDR	Clock and Data Recovery
5	CFR	Crest Factor Reduction
6	CISPR	International Special Committee on Radio Interference
7	CLB	Configurable Logic Block
8	CPU	Central Processing Unit
9	CRC	Cyclic Redundancy Check
10	CU	Centralized Unit as defined by 3GPP
11	DAC	Digital to Analog Converter
12	DDC	Digital Down Conversion
13	DDR	Double Data Rate
14	DEVCLK	Device Clock
15	DIMM	Dual-Inline-Memory-Modules
16	D2D	Die-to-Die Interface
17	DL	Downlink
18	DMA	Direct Memory Access
19	DPD	Digital Pre-Distortion
20	DPDK	Data Plane Development Kit
21	DPU	Digital Processing Unit
22	DSP	Digital Signal Processor
23	DU	Distributed Unit as defined by 3GPP
24	DUC	Digital Up Conversion
25	ECC	Error Correcting Code
26	eCPRI	evolved Common Public Radio Interface
27	EMC	Electro Magnetic Compatibility
28	ENIAC	Embedded NIC Inline Accelerator
29	EVM	Error Vector Magnitude

1	FCC	Federal Communications Commission
2	FEC	Forward Error Correction
3	FFT	Fast Fourier Transform
4	FH	Fronthaul
5	FHGW	Fronthaul Gateway
6	FHGW _x	Fronthaul gateway with no FH protocol translation, supporting an O-DU with split option x and an O-RU with split option x, with currently available options 6□6, 7-2□7-2 and 8□8
7		
8	FHGW _{x→y}	Fronthaul Gateway that can translate fronthaul protocol from an O-DU with split option x to an O-RU with split option y, with currently available option 7-2□8.
9		
10	FHHL	Full Height Half Length
11	FPGA	Field Programmable Gate Array
12	GbE	Gigabit Ethernet
13	GMC	Grand Master Clock
14	GNSS	Global Navigation Satellite System
15	GPIO	General Purpose Input Output
16	GPP	General Purpose Processor
17	GPS	Global Positioning System
18	GUI	Graphical User Interface
19	HAR	Hardware Architecture Requirement
20	HARQ	Hybrid Automatic Repeat request
21	HDD	Hard Disk Drive
22	HHHL	Half Height Half Length
23	HRD	Hardware Reference Design
24	HW	Hardware
25	I/O	Input/Output
26	I2C	Inter-Integrated Circuit (multi IC interface standard)
27	IEEE	Institute of Electrical and Electronics Engineers
28	IFFT	Inverse Fast Fourier Transform
29	IMC	Integrated Memory Controller

1	IMD	Inter Modulation Distortion
2	JTAG	Joint Test Action Group
3	KVM	Kernel-based Virtual Machine
4	L1	Layer 1
5	L2	Layer 2
6	LDPC	Low-Density Parity Codes
7	LNA	Low Noise Amplifier
8	LRDIMM	Load-Reduced Dual In-line Memory Module
9	LTE	Long Term Evolution
10	LUT	Look-up Table
11	MH	Midhaul
12	NEBS	Network Equipment-Building System
13	NFV	Network Functions Virtualization
14	NIC	Network Interface Controller
15	NR	New Radio
16	OAM	Operations, administration and management
17	O-CU	O-RAN Centralized Unit as defined by O-RAN
18	OCXO	Oven Controlled Crystal Oscillator
19	O-DU _x	A specific O-RAN Distributed Unit having fronthaul split option x where x may be 6, 7-2 (as defined by WG4) or 8.
20		
21	OMC	Outdoor Microcell
22	O-RU _x	A specific O-RAN Radio Unit having fronthaul split option x, where x is 6, 7-2 (as defined by WG4) or 8, and which is used in a configuration where the fronthaul interface is the same at the O-DU _x .
23		
24		
25	ORx	Observation Receiver
26	PA	Power Amplifier
27	PCH	Platform Controller Hub
28	PCIe	Peripheral Component Interface Express
29	PHC	Physical Hardware Clock
30	PHY	Physical Layer

1	PMBus	Power Management Bus
2	PPS	Pulse Per Second
3	PRACH	Physical Random-Access Channel
4	PTP	Precision Time Protocol
5	PWM	Pulse Width Modulation
6	QSFP	Quad Small Form-factor Pluggable
7	RAN	Radio Access Network
8	RDIMM	Registered Dual In-line Memory Module
9	RE	Resource Element
10	RF	Radio Frequency
11	RoE	Radio over Ethernet
12	RU	Radio Unit as defined by 3GPP
13	Rx	Receiver
14	SDFEC	Soft-Decision Forward Error Correction
15	SERDES	Serializer / De-serializer
16	SFP	Small Form-factor Pluggable
17	SFP+	Small Form-factor Pluggable plus
18	SMA	Sub-Miniature Version A (connector)
19	SoC	System On Chip
20	SPI	Serial Peripheral Interface
21	SSD	Solid State Drive
22	SyncE	Synchronous Ethernet
23	TCXO	Temperature Compensate Crystal Oscillator
24	TDD	Time Division Duplex
25	TDP	Thermal Design Power
26	TR	Technical Report
27	TR	Transmit Receive (switch)
28	TRx	Transceiver
29	TS	Technical Specification

1	Tx	Transmitter
2	UEFI	Unified Extensible Firmware Interface
3	UCIe	Universal Chiplet Interface Express
4	UL	Uplink
5	USB	Universal Serial Bus
6	UTC	Coordinate Universal Time
7	WG	Working Group

8 Chapter 2 Hardware Reference Design 1

9 This chapter describes a white box hardware reference design example for outdoor Microcell deployment
10 scenario. It includes O-CU and O-DU₇₋₂ for OMC deployment scenario.

11 2.1 O-CU Hardware Reference Design

12 The O-CU white box hardware is the platform that perform the O-CU function of upper L2 and L3. The
13 hardware systems specified in this document meet the computing, power and environmental requirements of
14 use cases configurations and feature sets of RAN physical node. These requirements are described in the early
15 hardware requirement specification as well as in the use cases document. The O-CU hardware includes the
16 chassis platform, mother board, peripheral devices and cooling devices. The mother board contains processing
17 unit, memory, the internal I/O interfaces, and external connection ports. The midhaul (MH) and backhaul (BH)
18 interface are used to carry the traffic between O-CU and O-DU₇₋₂ as well as O-CU and core network. The other
19 hardware functional components include: the storage for software, hardware and system debugging interfaces,
20 board management controller, just to name a few; the O-CU designer will make decision based on the specific
21 needs of the implementation.

22 The HW reference design of O-CU is the same as O-DU₇₋₂ except for the need of HW accelerator, thus detail
23 design will be described in O-DU₇₋₂ section 2.2.

24 2.2 O-DU₇₋₂ Hardware Reference Design

25 The O-DU₇₋₂ white box hardware is the platform that performs the O- DU₇₋₂ functions such as upper L1 and
26 lower L2 functions. The hardware systems specified in this document meet the computing, power and
27 environmental requirements of use case's configurations and feature sets of RAN physical node. These
28 requirements are described in the early hardware requirement specification as well as in the use cases document.
29 The O-DU₇₋₂ hardware includes the chassis platform, mother board, peripheral devices and cooling devices.
30 The mother board contains processing unit, memory, the internal I/O interfaces, and external connection ports.
31 The fronthaul and backhaul interface are used to carry the traffic between O-RU₇₋₂ / FHGW₇₋₂ / FHGW_{7-2→8}
32 and O-DU₇₋₂ as well as O-CU and O-DU₇₋₂. The O-DU₇₋₂ design may also provide an interface for hardware
33 accelerator if that option is preferred. The other hardware functional components include: the storage for

software, hardware and system debugging interfaces, board management controller, just to name a few; the O-DU₇₋₂ designer will make decision based on the specific needs of the implementation.

Note that the O-DU₇₋₂ HW reference design is also feasible for O-CU and integrated O-CU / O-DU₇₋₂.

2.2.1 O-DU₇₋₂ High-Level Functional Block Diagram

Figure 2.2.1-1 shows the major functional blocks of O-DU₇₋₂. The digital processing unit handles the baseband processing workload. To make the processing more efficient, an accelerator can be used to assist with the baseband workload processing. The memory devices include the random-access memory (RAM) for temporary storage of data while flash memory is used for codes and logs. The storage device is for persistent storage. The external network cards can be used for fronthaul or backhaul connection. The baseboard management controller (BMC) is a microcontroller which monitors hardware operation on motherboard. The clock circuits provide digital processing unit with required clock signals.

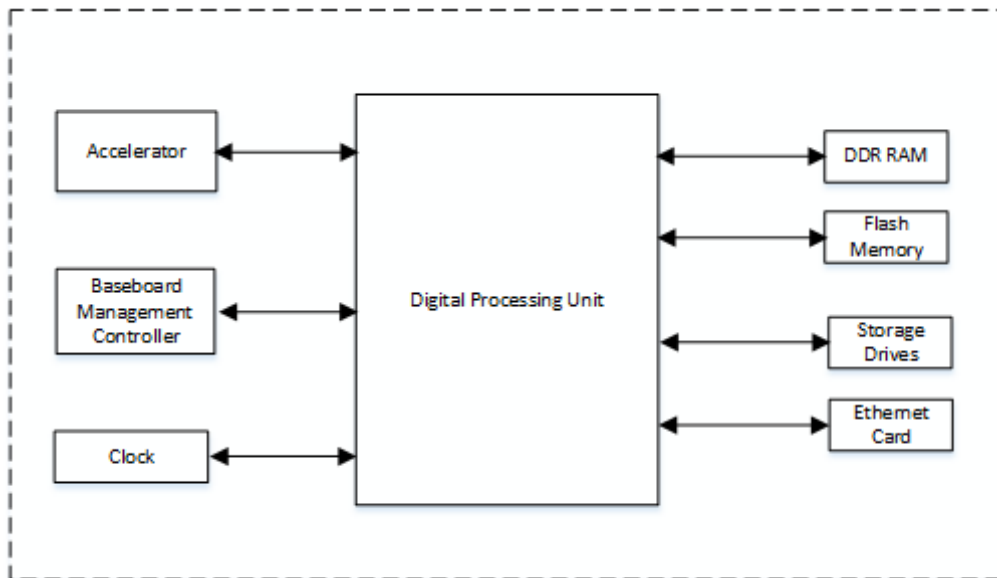


Figure 2.2.1-1 O-DU₇₋₂ Functional Block Diagram

2.2.2 O-DU₇₋₂ Hardware Design Description

Figure 2.2.2-1 describes the various components and connections inside the O-DU₇₋₂ white box.

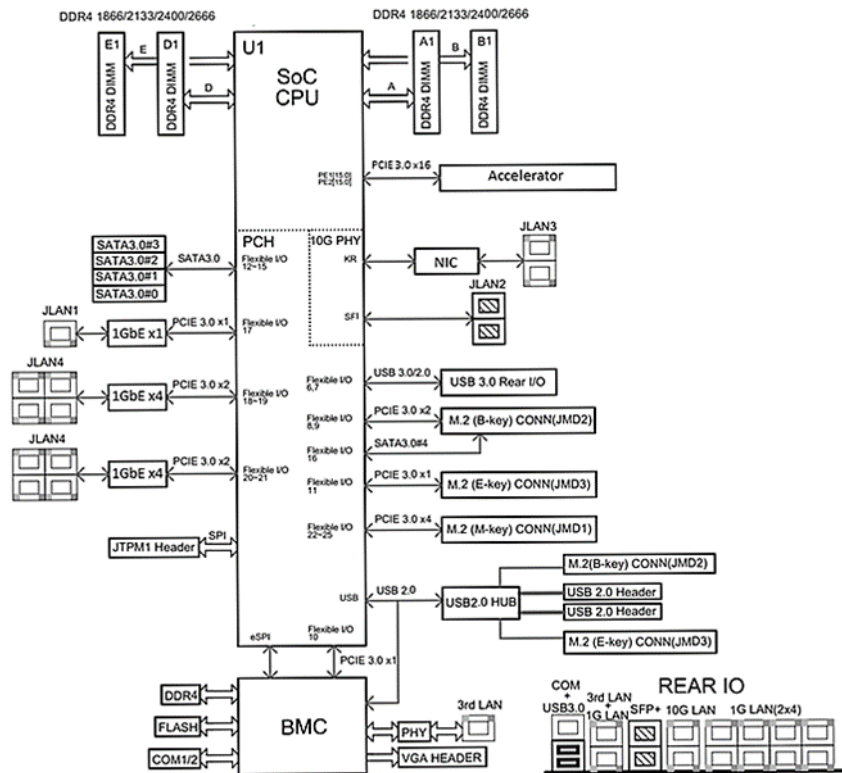


Figure 2.2.2-1 O-DU₇₋₂ Hardware Block Diagram

As described in the previous section, the O-DU₇₋₂ hardware can be implemented with difference design choices. Here, we use a SoC design-based as an example which performs most of the workload for O-DU₇₋₂. The accelerator can be used to perform other functions based on the overall performance requirement. Several Ethernet controllers are also used for front haul link, back haul link and remote console control connection. The other parts include: RAM, flash memory, and hard drive storage. The JTAG and USB ports are provided for hardware debug and local connection if needed. Finally, BMC block is mainly responsible for monitoring the hardware status of the motherboard.

2.2.3 O-DU₇₋₂ Hardware Components

In this section, we describe the details of the O-DU₇₋₂ hardware component's requirements, their features and parameters. The component selection is based on the use case requirements which are listed in the hardware architecture and requirements document [6].

2.2.3.1 Digital Processing Unit

This example of the digital processing unit in O-DU₇₋₂ is based on the General-Purpose Processor (GPP).

2.2.3.1.1 Digital Processing Unit Requirements

The GPP requirements are listed in Table 2.2.3-1.

Table 2.2.3-1 Processor Feature List

Item Name	Description
# of Cores	16
# of Threads	32
Base Frequency	2.20 GHz
Max Turbo Frequency	3.00 GHz
Cache	22 MB
Thermal Design Power (TDP)	100W
Max Memory Size (dependent on memory type)	512 GB
Memory Types	DDR4
Max # of Memory Channels	4

2.2.3.1.1.1 Interfaces

The interface specifications on the main board are described in the sections below.

2.2.3.1.1.1.1 Memory Channel Interfaces

The system memory capacity, type and related information are described in Table 2.2.3-2.

Table 2.2.3-2 Memory Channel Feature List

Item Name	Description
Memory Types	DDR4
# of Memory Channels	4
ECC LRDIMM	Up to 512GB
ECC RDIMM	Up to 256GB
Memory Speed	2666/2400/2133MHz
DIMM Sizes	128GB, 64GB, 32GB, 16GB
Memory Voltage	1.2 V

2.2.3.1.1.1.2 PCIe

PCIe Gen 3 should be supported by the processor. There are total of 32 PCIe lanes with 128 Gb/s bandwidth. The 32 PCIe lanes can be divided into two x16 slots by using a riser card.

2.2.3.1.1.1.3 Ethernet

The system should be capable to offer aggregated 48 Gb/s Ethernet bandwidth. The breakout the ports are discussed in later section. When higher Ethernet bandwidth required, an Ethernet card can be installed in one of the PCIe slot.

2.2.3.1.2 Digital Processing Unit Design

The digital processing unit is a system-on-a-chip (SoC) device which is a 64-bit multi-core server class processor. This SoC includes an integrated Platform Controller Hub (PCH), integrated high-speed I/O, Integrated Memory Controllers (IMC), and four integrated 10 Gigabit Ethernet ports.

The SoC supports 512-bit wide vector processing instruction set. It also supports hardware virtualization to enable dynamic provisioning of services as communication service providers extend network functions virtualization (NFV). Figure 2.2.2-1 shows the major functional blocks of the digital processing unit.

2.2.3.2 Hardware Accelerator

Hardware accelerators can be used in O-DU₇₋₂ to offload computationally intensive functions and to optimize the performance under varying traffic and loading conditions. While the hardware acceleration functional requirements and implementation are system designer's choice; however, the O-DU₇₋₂ is required to meet the minimum system performance requirements under various loading conditions and deployment scenarios. In most cases, a Field Programmable Gate Array (FPGA) or Application Specific Integrated Circuit (ASIC) based PCIe card can be used to optimize the system performance. The FPGA(s) are part of a Network Interface Controller (NIC) that further provides connectivity services.

2.2.3.2.1 Accelerator Design Solution 1

Channel coding for Low-Density Parity Codes (LDPC) and fronthaul compression requires a significant amount of bit level processing and is well suited to a fine-grained FPGA architecture and/or low cost/power structured ASIC. Options include:

- Virtualization: Multi-channel DMA, Virtual machine integration, load balancing functions
- Look-aside FEC: LDPC (de)coding, Polar (de)coding, Rate (De)Matcher, (De) Interleaver, Cyclic Redundancy Check (CRC), Hybrid Automatic Repeat request (HARQ) retransmission
- Bump-in-wire Fronthaul: compression / decompression for latency and bandwidth reduction

2.2.3.2.1.1 Accelerator Requirements

Hardware and firmware requirements for this accelerator design are given in Table 2.2.3-3 and Table 2.2.3-4, respectively.

Table 2.2.3-3 Accelerator Hardware Feature List

Item Name	Description
PCIe (Interface with digital processing unit)	Gen4 x16 (and lower)
Form factor	FHHL
Connectivity	2x QSFP28/56
FPGA	Logic Elements: 1437K M20K Memory: 139Mb Quad A53 Hard Processor Sub-system
NIC Device	100Gb xHAUL for FH, BH & MH traffic shaping. Optional FPGA co-processing.
DDR Main	8+8+1GB DDR4
Flash (FPGA images)	>=1 Gbit
BMC	Telemetry, Security, remote upgrade
Clocking	For O-RAN C1, C2, C3 & C4
Fronthaul	eCPRI, RoE IEEE1914.3, O-RAN WG4
GPS	SMA for 1 PPS & 10MHz (in/out)
Operating Temperature (ambient)	NEBS Compliant
Power	<75W (without optional NIC device)
Clock Accuracy	Low-Jitter, configurable clock ranging from 10MHz to 750MHz. Option for OCXO (TCXO as standard)

Table 2.2.3-4 Accelerator Firmware Feature List

Item Name	Description
Remote system upgrade	Securely upgradable FPGA flash image
Queuing	64 Queues supported equally split between UL & DL.
LDPC Acceleration	NR LDPC Encoding with, interleaving and rate-matching. NR LDPC Decoding with sub-block de-interleaving function of reverse rate matching. Early Termination, CRC attachment and HARQ buffering. 5G Throughput: DL 14.8Gbps, UL 3.2Gbps
Load Balancing (channel coding)	Load balancer distributes the pending encoder/decoder requests to encoder/decoders
Descriptor Format (channel coding)	Code block-based interface. Software enablement by BBDEV API (DPDK) [7], [8]
Fronthaul Compression	In-line compression/decompression for Mu-Law, block-floating point and quantization according to the O-RAN WG4 specification.
Open programmable acceleration environment	Support for: - FPGA Flashing upgrade - Firmware version reporting - PCIe diagnostics - Ethernet diagnostics - Temperature and voltage telemetry information

2.2.3.2.1.2 Accelerator Design

Figure 2.2.3-1 illustrates Accelerator Design 1 without optional NIC Device. Figure 2.2.3-2 illustrates the Accelerator Design 1 with optional NIC Device.

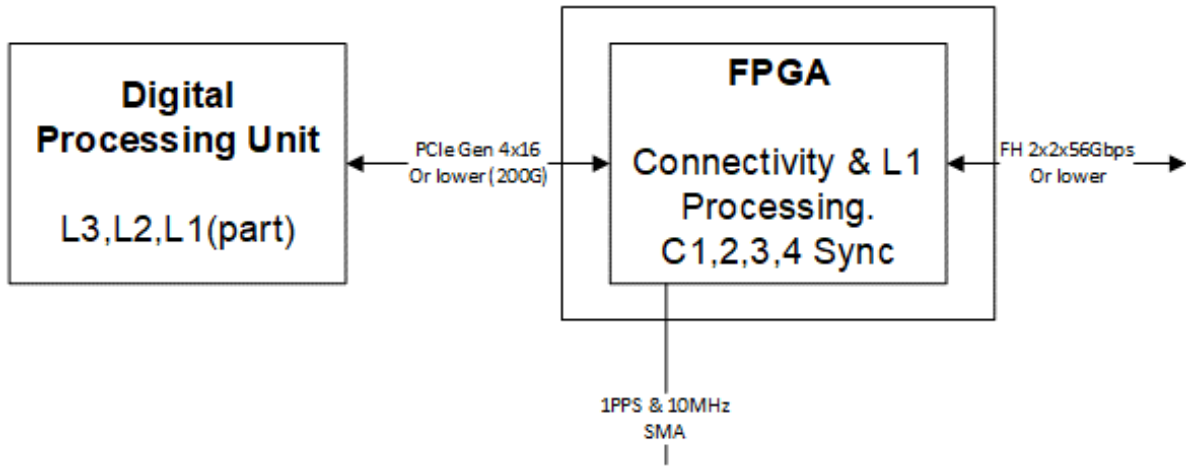


Figure 2.2.3-1 Accelerator Design 1 without optional NIC Device

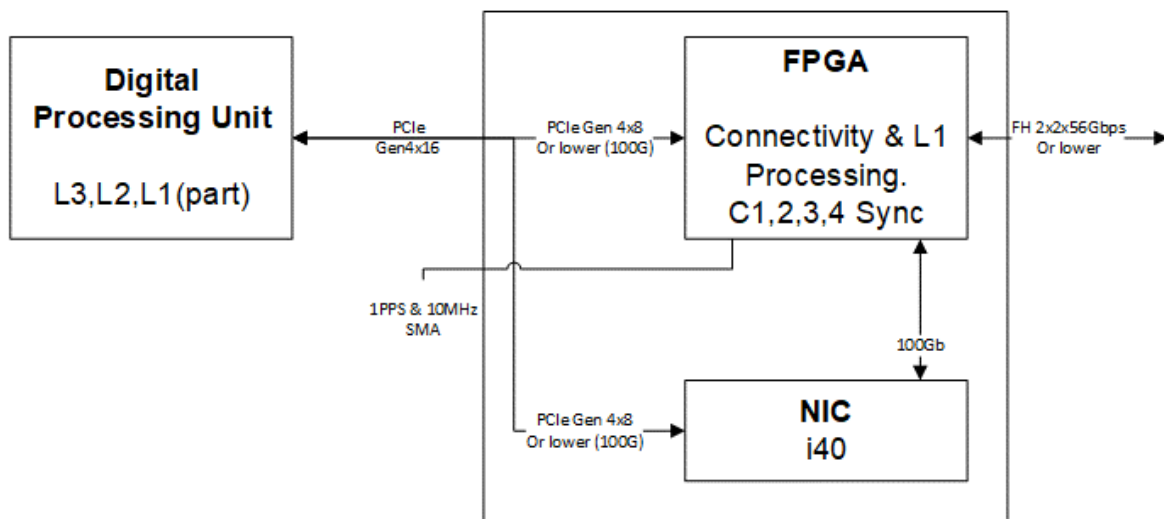


Figure 2.2.3-2 Accelerator Design 1 with optional NIC Device

2.2.3.2.2 Accelerator Design Solution 2

Channel coding for LDPC and fronthaul compression requires a significant amount of bit level processing and is well suited to a fine-grained structured ASIC. Features include:

- Virtualization: Multi-channel DMA, Virtual machine integration, load balancing functions
- Look-aside FEC: Turbo (de)coding, LDPC (de)coding, Rate (De)Matcher, (De) Interleaver, CRC, HARQ retransmission & (de) interleaver.

2.2.3.2.2.1 Accelerator Requirement

Accelerator Hardware and Firmware features are listed in Table 2.2.3-5 and Table 2.2.3-6, respectively.

Table 2.2.3-5 Accelerator Hardware Feature List

Item Name	Description
PCIe	Gen3 x16
Form factor	HHHL
DDR	DDR4 (64-bit +ECC), 2667Mbps Interface for HARQ buffering
Board Management Controller	Telemetry, Security.
Power	<35W

Table 2.2.3-6 Accelerator Firmware Feature List

Item Name	Description
Queuing	64 Queues supported equally split between UL & DL.
LDPC Acceleration	NR LDPC Encoding with, interleaving and rate-matching. NR LDPC Decoding with sub-block de-interleaving function of reverse rate matching. Early Termination, CRC attachment and HARQ buffering. 5G Throughput: DL 23Gbps, UL 8Gbps
Load Balancing (channel coding)	Load balancer distributes the pending encoder/decoder requests to encoder/decoders
Descriptor Format (channel coding)	Code block and transport block-based interface. Software enablement by BBDEV API (DPDK) [7], [8]

2.2.3.2.2.2 Accelerator Design

The following diagram shows the structured ASIC based accelerator design.

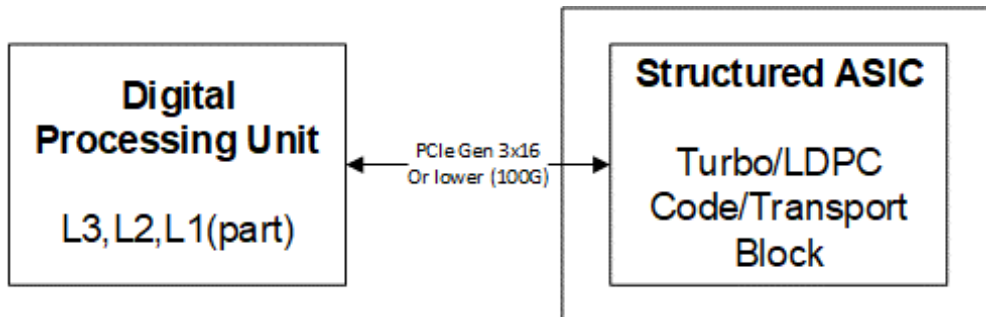


Figure 2.2.3-3 Structured ASIC Accelerator Design

2.2.3.2.3 Accelerator Design Solution 3

The O-DU₇₋₂ system is typically implemented using a multi-core processor and one or more hardware accelerators. Parts of O-DU₇₋₂ protocol stack can be implemented in software running on the multi-core processors and the computationally intensive L1 and L2 functions can be offloaded to an FPGA-based hardware accelerator. This accelerator comprises two FPGAs for L1 offload and fronthaul connectivity in lookaside mode. The accelerator hardware and firmware features are listed in Table 2.2.3-7 and Table 2.2.3-8.

2.2.3.2.3.1 Accelerator Hardware Features

Table 2.2.3-7 Accelerator Hardware Features

Item Name	Description	
SoC Resources	FPGA 1 System Logic cells - 930K CLB LUT - 425K SDFEC -8 DSP Slices - 4,272 BRAM - 38.0Mb URAM - 22.5Mb	FPGA 2 System Logic cells - 1,143K CLB LUT - 523K CLB Flip-Flops -1,045K DSP Slices - 1,968 BRAM - 34.6Mb URAM - 36.0Mb
Form Factor	FHHL PCIe Form Factor	
PCIe Interface	Gen 3 x16 with Bifurcation	
On Board Memory	FPGA1 Total Capacity 4 GB in Programmable Logic, upgradeable to 8GB Total Capacity 2 GB in Programmable Subsystem, upgradeable to 4GB	FPGA2 Total Capacity 4 GB in Programmable Logic, upgradeable to 8GB Total Capacity 2 GB in Programmable Subsystem, upgradeable to 4GB
Network Interface(s)	2xSFP28 optical interfaces to FPGA2 (User Configurable, includes 10/25 Ethernet)	
Other External Interface(s)	Micro USB for JTAG support (FPGA programming and debug) and access to BMC	
Graphical User interface	GUI for monitoring the basic board parameters, monitoring temperature alerts, firmware upgrades for BMC	
Board Management Controller	Telemetry, Security, Remote Upgrade	
Clocking	O-RAN C1, C2, C3, C4	
Fronthaul	eCPRI, RoE IEEE1914.3, O-RAN WG4 FH	
GPS	GPS SMA for 1 PPS (In/Out)	
Operating Temperature	NEBS Compliant	
Power	< 75 W	
Clocking Options	Low-Jitter, configurable clock ranging from 10MHz to 750MHz 1 PPS input and output with assembly option for OCXO and TCXO	

2.2.3.2.3.2 Hardware Accelerator Firmware Features

Table 2.2.3-8 Hardware Accelerator Firmware Features

Item Name	Description
Remote System Upgrade	Securely upgradable FPGA flash image
L1 Acceleration	5G NR LDPC encoding/decoding with interleaving/de-interleaving and rate-matching/rate-de-matching along with early termination, CRC attachment, and HARQ management
Hardware-Software APIs	Fully compliant with DPDK/BBDEV APIs
Fronthaul Protocols/Compression	Fully compliant with O-RAN WG4 M, C,U & S-planes Specifications
Open Programmable Acceleration Environment	Support for: - FPGA flashing upgrade - Firmware version reporting - PCIe diagnostics - Ethernet diagnostics - Temperature and voltage telemetry information

The hardware accelerator supports x86 or non-x86-based processors. Figure 2.2.3-4 illustrates the two-chip acceleration architecture comprising two FPGAs with multi-lane PCIe interfaces toward the CPU and external connectivity toward O-RU₇₋₂(s) via eCPRI and O-CU(s) through GbE connectivity. The example architecture further depicts multi-lane Gen3 PCIe interfaces between each FPGA and the CPU. The FPGAs communicate through high-bandwidth Ethernet (GbE) transport.

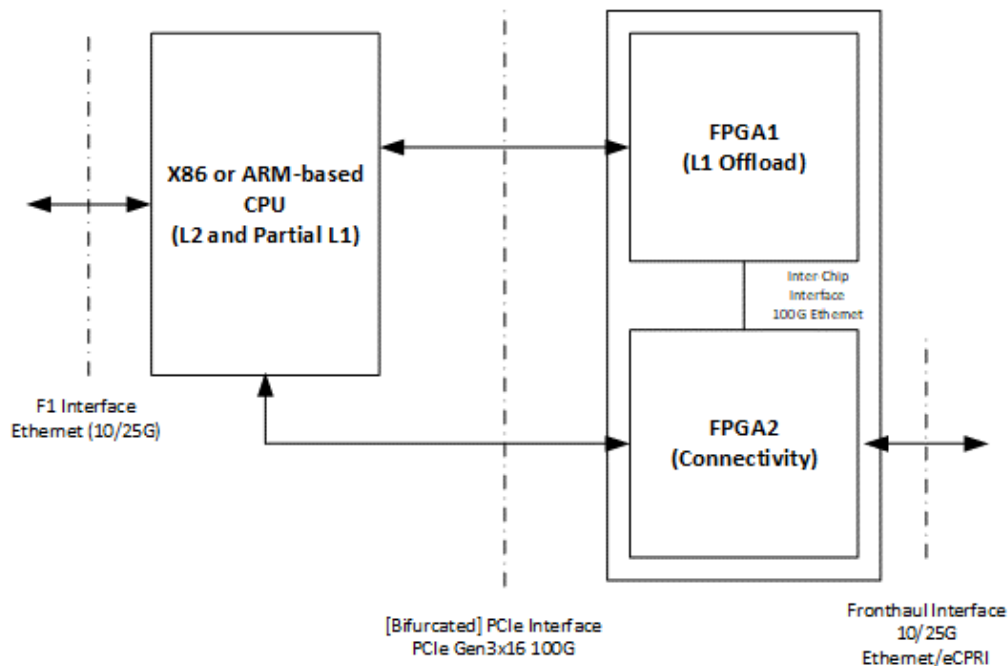


Figure 2.2.3-4 Dual-chip FPGA-based Hardware Acceleration in O-DU₇₋₂

2.2.3.2.4 Accelerator Design Solution 4

The O-DU₇₋₂ system is typically implemented using a multi-core processor (CPU) and one or more hardware accelerators. Parts of O-DU₇₋₂ protocol stack can be implemented in software running on the multi-core processors and the computationally intensive L1 and L2 functions can be offloaded to an FPGA-based hardware accelerator. This accelerator comprises a single FPGA for L1 offload in lookaside mode. The accelerator hardware and firmware features are listed in Table 2.2.3-9 and Table 2.2.3-10.

2.2.3.2.4.1 Accelerator Hardware Features

Table 2.2.3-9 Accelerator Hardware Features

Item Name	Description
SoC Resources	System Logic cells - 930K CLB LUT - 425K SDFEC -8 DSP Slices - 4,272 BRAM - 38.0Mb URAM - 22.5Mb
Form Factor	HHHL PCIe Form Factor
PCIe Interface	Gen 3 x16 (Gen4 x8)
On Board Memory	Total Capacity 4 GB in Programmable Logic, upgradeable to 8GB Total Capacity 2 GB in Processor Subsystem, upgradeable to 4GB
Other External Interface(s)	Micro USB for JTAG support (FPGA programming and debug) and access to BMC
Graphical User interface	GUI for monitoring the basic board parameters, monitoring temperature alerts, firmware upgrades for BMC
Board Management Controller	Telemetry, Security, Remote Upgrade
Operating Temperature	NEBS Compliant
Power	< 35 W

2.2.3.2.4.2 Hardware Accelerator Firmware Features

Table 2.2.3-10 Hardware Accelerator Firmware Features

Item Name	Description
Remote System Upgrade	Securely upgradable FPGA flash image
L1 Acceleration	5G NR LDPC encoding/decoding with interleaving/de-interleaving and rate-matching/rate-de-matching along with early termination, CRC attachment, and HARQ management
Hardware-Software APIs	Fully compliant with DPDK/BBDEV APIs
Open Programmable Acceleration Environment	Support for: - FPGA flashing upgrade - Firmware version reporting - PCIe diagnostics - Ethernet diagnostics - Temperature and voltage telemetry information

The hardware accelerator supports x86 or non-x86-based processors. Figure 2.2.3-5 illustrates the single-chip acceleration architecture comprising one FPGA with Gen3 x16 or Gen4 x8 PCIe interfaces toward the CPU.

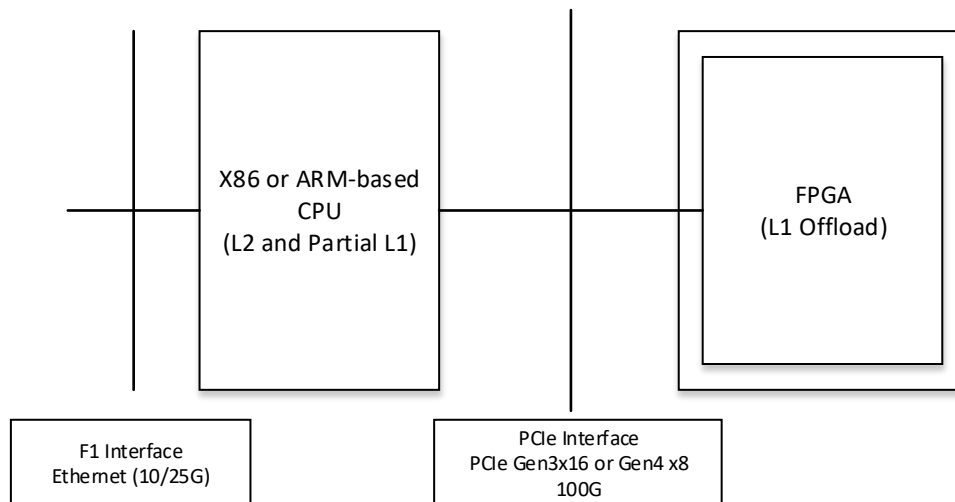


Figure 2.2.3-5 Single-chip FPGA-based Hardware Acceleration in O-DU₇₋₂

2.2.3.2.5 Accelerator Design Solution 5

This design includes an Embedded NIC Inline Accelerator (ENIAC) with an ASIC that is capable of providing NIC functions. One option supports a high-capacity O-DU using multiple PCIe cards in standard racks and aligns with the FAPI interface to maximize software reuse. The second option supports a high-capacity O-DU using an Ethernet-based backplane interconnect and aligns with the nFAPI interface to maximize software reuse. ENIAC unloads L1 from the host, allowing it to focus on L2.

2.2.3.2.5.1 Accelerator Requirements

Hardware and firmware requirements for this accelerator design are given in Table 2.2.3-11 and Table 2.2.3-12, respectively.

Table 2.2.3-11 Accelerator Hardware Feature List]

Item Name	Description
PCIe (Interface with digital processing unit)	Gen4 x16 (and lower)
Form factor	FHHL
Connectivity	QSFP28/SFP
NIC Device	100Gb xHAUL for FH, BH & MH traffic shaping.
DDR Main	Min 4 GB DDR4
Flash (FPGA images)	>=1 Gbit
BMC	Telemetry, Security, remote upgrade
Clocking	For O-RAN C1, C2, C3 & C4
Fronthaul	eCPRI, RoE IEEE1914.3, O-RAN WG4
GPS	SMA for 1 PPS & 10MHz (in/out)
Operating Temperature (ambient)	NEBS Compliant
Power	<75W
Clock Accuracy	Low-Jitter, configurable clock ranging from 10MHz to 750MHz. Option for OCXO (TCXO as standard)

Table 2.2.3-12 Accelerator Firmware Feature List

Item Name	Description
Remote system upgrade	Securely upgradable image
Queuing	Command queuing at FAPI interface
LDPC Acceleration	Rel15 compliant
Fronthaul Compression	In-line compression/decompression for Mu-Law, block-floating point and quantization according to the O-RAN WG4 specification.
Open programmable acceleration environment	Support for: - Flashing upgrade - Firmware version reporting - PCIe diagnostics - Ethernet diagnostics - Temperature and voltage telemetry information.

2.2.3.2.5.2 Accelerator Design

Figure 2.2.3-6 illustrates Accelerator Design 5 with integrated NIC and PCIe connectivity to L2. Figure 2.2.3-7 illustrates Accelerator Design 5 with integrated NIC and Ethernet connectivity to L2.

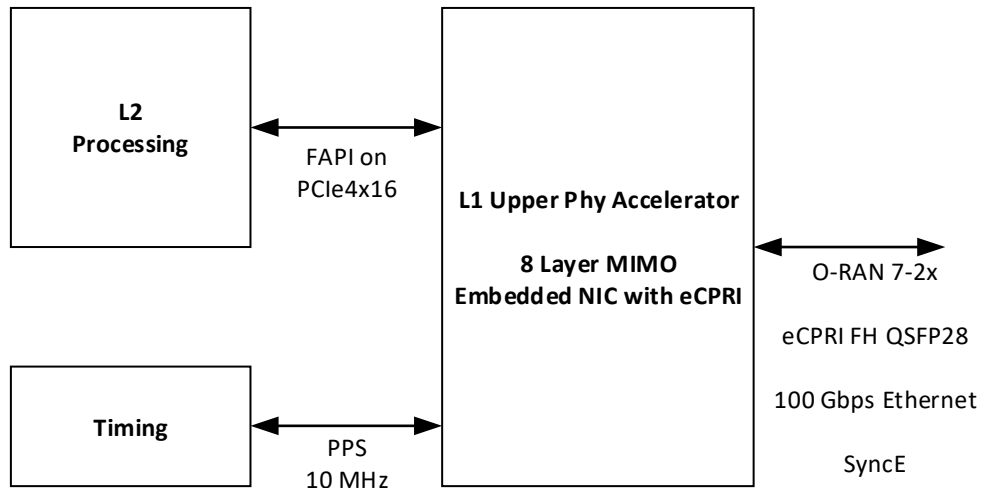


Figure 2.2.3-6 Accelerator Design Using PCIe

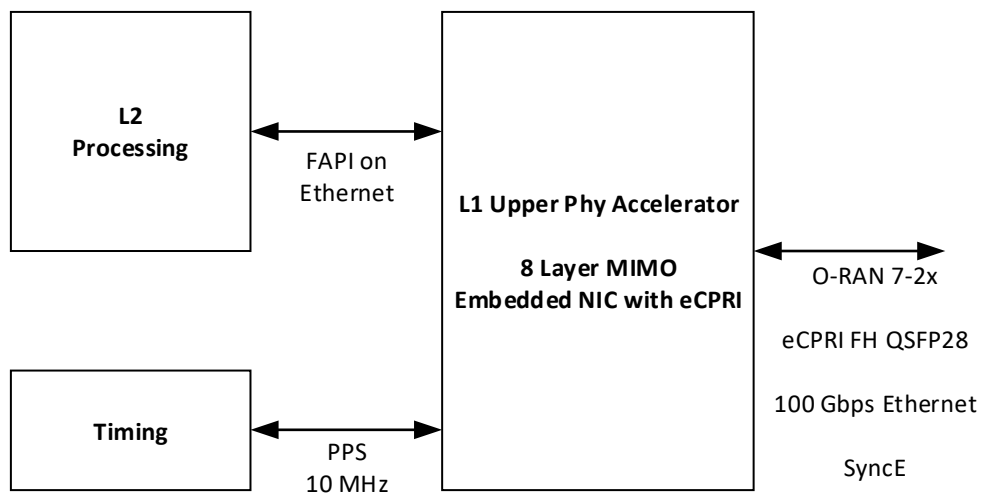


Figure 2.2.3-7 Accelerator Design Using Ethernet

2.2.3.3 Baseboard Management Controller

BMC is used to perform hardware power control (power on, power off and power cycle), monitor hardware status (temperatures, voltages, etc), monitor Basic I/O System (BIOS)/ Unified Extensible Firmware Interface (UEFI) firmware status, and log system events. It provides remote access via shared or dedicated NIC. System user can do console access via serial or physical/Kernel-based Virtual Machine (KVM). The BMC has dedicated RAM and flash memory. It provides access via serial port or Ethernet port. Figure 2.2.2-1 describes the BMC connections with related components.

2.2.4 Synchronization and Timing

2.2.4.1 Synchronization and Timing Design 1

This section describes the synchronization and timing mechanism that is used in the O-DU₇₋₂.

2.2.4.2 Hardware Requirements

The O-DU₇₋₂ shall support following timing synchronization methods:

1. GNSS Synchronization
2. Ethernet based IEEE1588V2 Synchronization
3. GNSS and IEEE1588 switching

2.2.4.3 Hardware Design

Depending on the timing distribution topologies used, the O-DU₇₋₂ system clock is able to synchronize with the Grand Master Clock (GMC) using IEEE1588 via either the front haul NIC or backhaul NIC or synchronizing timing using Global Navigation Satellite System (GNSS). In the case of IEEE1588, the Physical Hardware Clock (PHC) within the NIC is synchronized with the GMC first, then the O-DU₇₋₂ system clock is synchronized with the PHC. The O-DU₇₋₂ is also capable to provide a clock to the O-RU₇₋₂ via front haul if needed. When GNSS becomes available to O-DU₇₋₂, it will be able to synchronize the system clock to Coordinated Universal Time (UTC). Figure 2 7 outlines the O-DU₇₋₂ timing synchronization mechanisms.

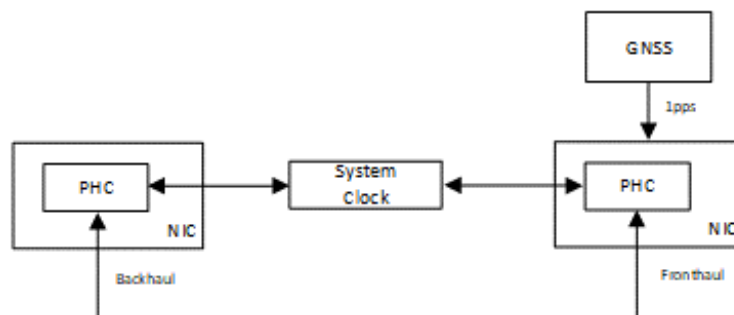


Figure 2.2.4-1 O-DU Timing Synchronization

2.2.5 External Interface Ports

The external interfaces of O-DU₇₋₂ are described below.

2.2.5.1 Hardware Requirements

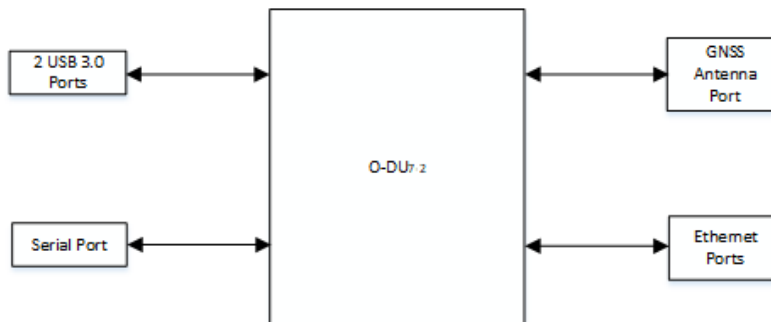
The following table shows the external ports or slots that the system provides.

Table 2.2.5-1 External Port List

Item Name	Description
Ethernet	Octave Gigabit Ethernet LAN connectors Dual 10GbE Base-T Ethernet connectors Dual 10GbE SFP+ Faber Ethernet connectors 2x100G QSFP28 or 2x25G SFP28
USB	2 USB 3.0 ports
Serial Port	1 COM port via RJ45
Antenna port	1 SMA connector for GNSS

2.2.5.2 Hardware Design

The digital processing unit is a SoC device which provides the external ports described in the hardware requirement section. The system includes 2 USB 3.0 ports, and the serial RS232 port that can be used for Console Redirection, e.g. Out-of-Band Management. The system provides eight 1Gbps and four 10Gbps Ethernet ports. There are two or four 25G eCPRI ports in system depends on the accelerator card used. The system also provides a RF interface to connect GNSS antenna. The following diagram outlines the external interfaces that supported.


Figure 2.2.5-1 External Interface Reference Design

2.2.6 O-DU Firmware

BIOS and BMC firmware are needed in the system and shall be installed.

2.2.7 Mechanical

The mechanical design for mother board, chassis, and cooling are listed in this section.

2.2.7.1 Mother Board

The mechanical layout of the mother board shows the location of major components and interface ports. The following diagram also provides the dimension of the board.

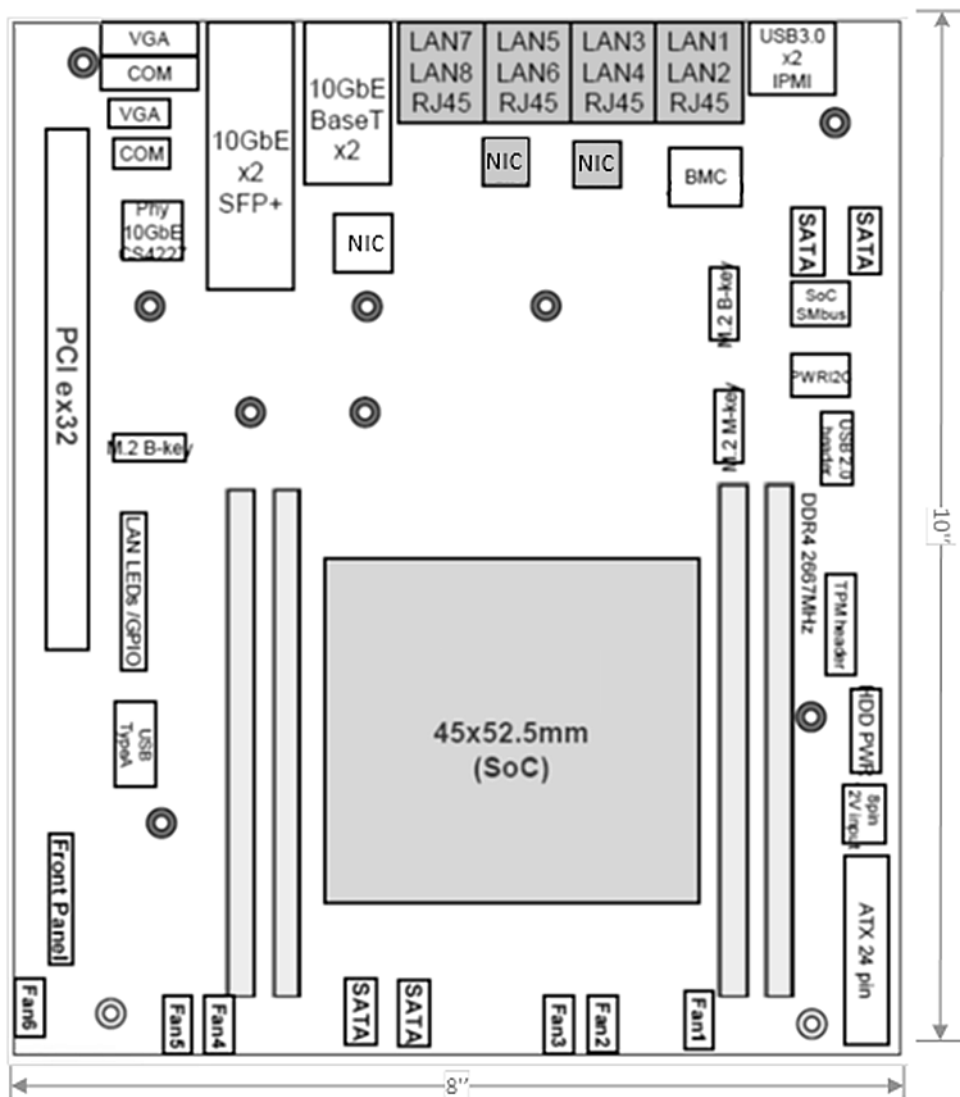


Figure 2.2.7-1 Mother Board Layout Diagram

2.2.7.2 Chassis

The 1U rack mount chassis contains the layout of the power supply, Solid State Drive (SSD) and fans. The chassis dimension is showed in following figure.

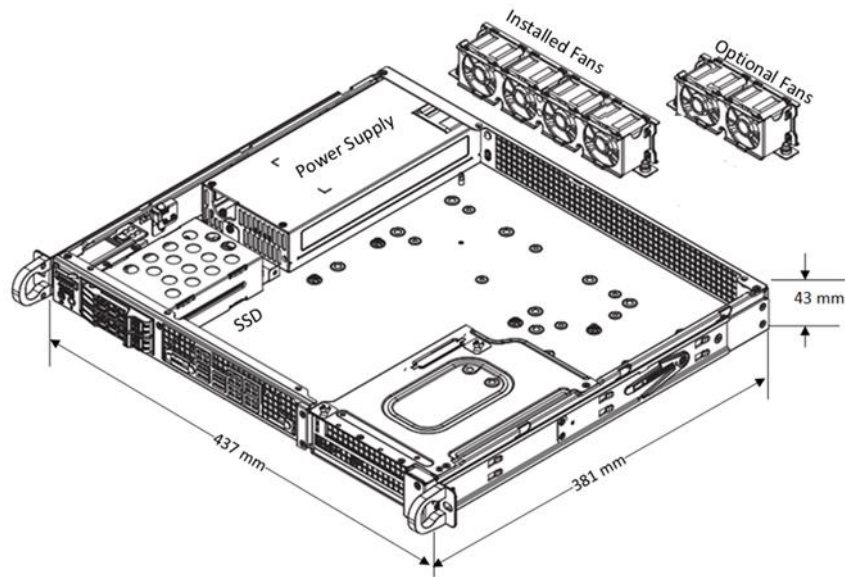


Figure 2.2.7-2 Chassis Mechanical Diagram

2.2.7.3 Cooling

The system installs 4x 40x28mm PWM fans for the cooling. Up to 6 fans can be installed if needed.

2.2.8 Power Unit

In a fully loaded system with two PCIe slots populated with 75W each, the system power consumption should be less than 400W. The total system power requirement shall be kept less than 80% of the power supply capacity.

2.2.8.1 Hardware Requirements

The O-DU requires a 500W high-efficiency power supply with Power Management Bus (PMBus) 1.2 capability. The required power input range and output power rails are listed in the table below.

Table 2.2.8-1 Power Supply Rail Requirements

Item Name	Description
AC Input	100-240V, 50-60Hz, 6.6A max
DC Output	+3.3V: 12A +5V: 15A +5V standby: 3A +12V: 41A -12V: 0.2A

2.2.8.2 Hardware Design

The O-DU₇₋₂ chassis design includes one 500W power supply unit. The power supply unit is auto-switching capable, which enables it to automatically sense and operate at a 100v to 240v input voltage. The power supply unit features are listed in the following table.

Table 2.2.8-2 Power Supply Unit Feature List

Item Name	Description
Output connectors	24pin/8pin/4+4pin/HDD/I2C
Dimension (W x D x H)	3.9 x 7.1 x 1.6 inch
Maximum Output Power	+3.3V: 12A +5V: 15A +12V: 41A -12V: 0.2A +5Vsb: 3A
Rated Input Voltage/Current	100-240Vac / 6.6A max
Rated Input Frequency	50-60HZ
Inrush current	Less than 30A

2.2.9 Thermal

Active cooling with up to 6 fans is integrated in the chassis.

The hardware acceleration cards described in 2.2.3.2 use passive cooling and a custom heatsink and is equipped with temperature sensors. It is designed to operate in temperatures ranging from -5°C to +55°C.

2.2.10 Environmental and Regulations

The O-DU₇₋₂ hardware system is RoHS Compliant. The power supply unit is EMC FCC/CISPR Class B compliant. Table 2 10 lists the environmental related features and parameters.

Table 2.2.10-1 Environmental Features

Item Name	Description
Operating Temperature	-5°C to +55°C
Non-operating Temperature	-40°C to 70°C
Operating Relative Humidity	8% to 90% (non-condensing)
Non-operating Relative Humidity	5% to 95% (non-condensing)

The hardware accelerator described in 2.2.3.2 is designed to operate in indoor environments and in temperatures ranging from -5°C to +55°C.

2.3 O-RU₇₋₂ Hardware Reference Design

This chapter defines the hardware reference design for the Outdoor Microcell O-RU₇₋₂. This is defined by a 7-2 interface to the O-DU on one side and up to 8T8R on the antenna port. Because this design is up to 10W per channel of conducted power, the external power is supplied by either AC or a standard 48V telecom source.

2.3.1 O-RU₇₋₂ High-Level Functional Block Diagram

Figure 2.3.1-1 provides a high-level functional block diagram depicting the major HW/SW components. It also highlights the internal/external interfaces that are required. This document shows how to implement the system defined by the OMC-HAR [6] document.

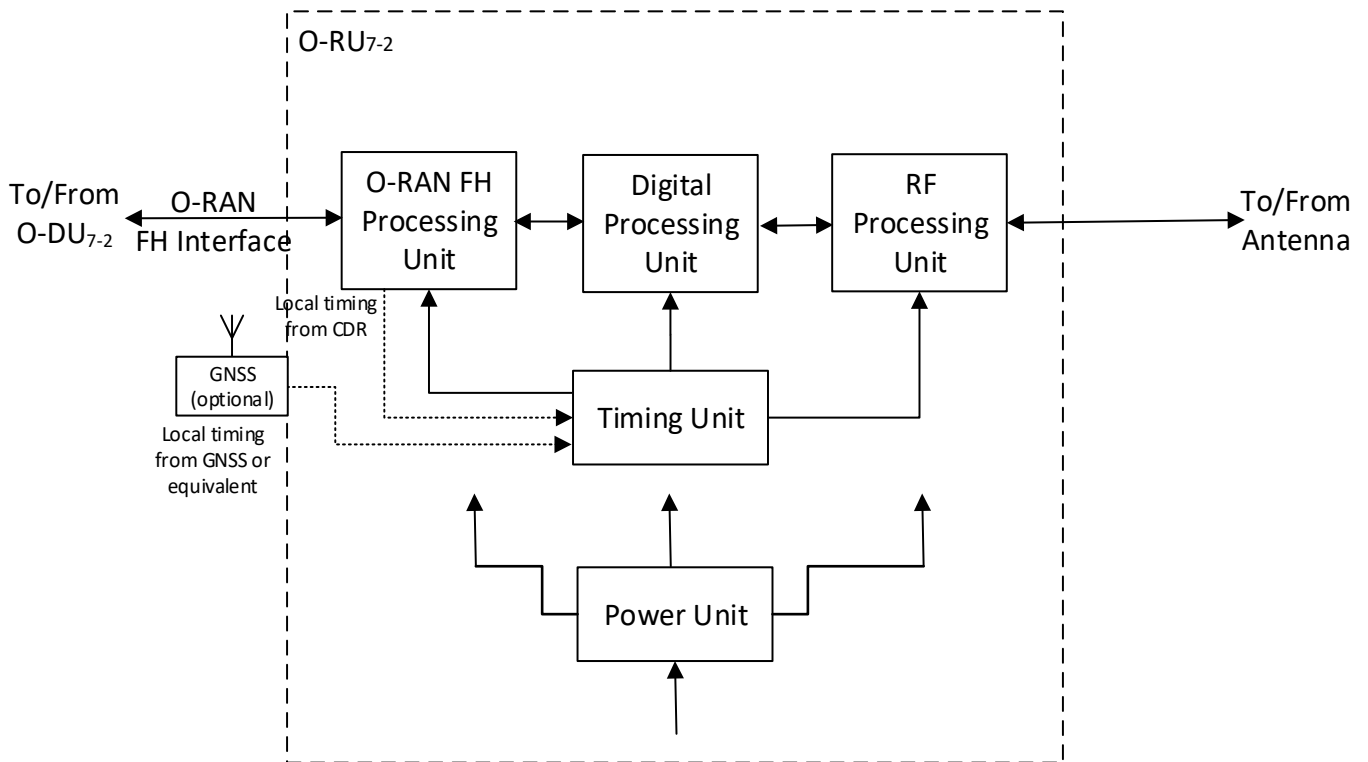


Figure 2.3.1-1 High-Level Functional Block Diagram

2.3.2 O-RU₇₋₂ Hardware Components

Figure 2.3.2-1 shows a 4T4R implementation and Figure 2.3.2-2 shows an 8T8R implementation. In each of these diagrams, the Digital Processing Unit is further detailed in 2.2.3.1. Items under the umbrella of RF Processing Unit, including the Transceiver, RFFE and other RF items are reviewed in additional detail in 2.3.2.2. Clock and Synchronization are reviewed in 2.3.2.3. The Power Unit is detailed in 2.3.6.

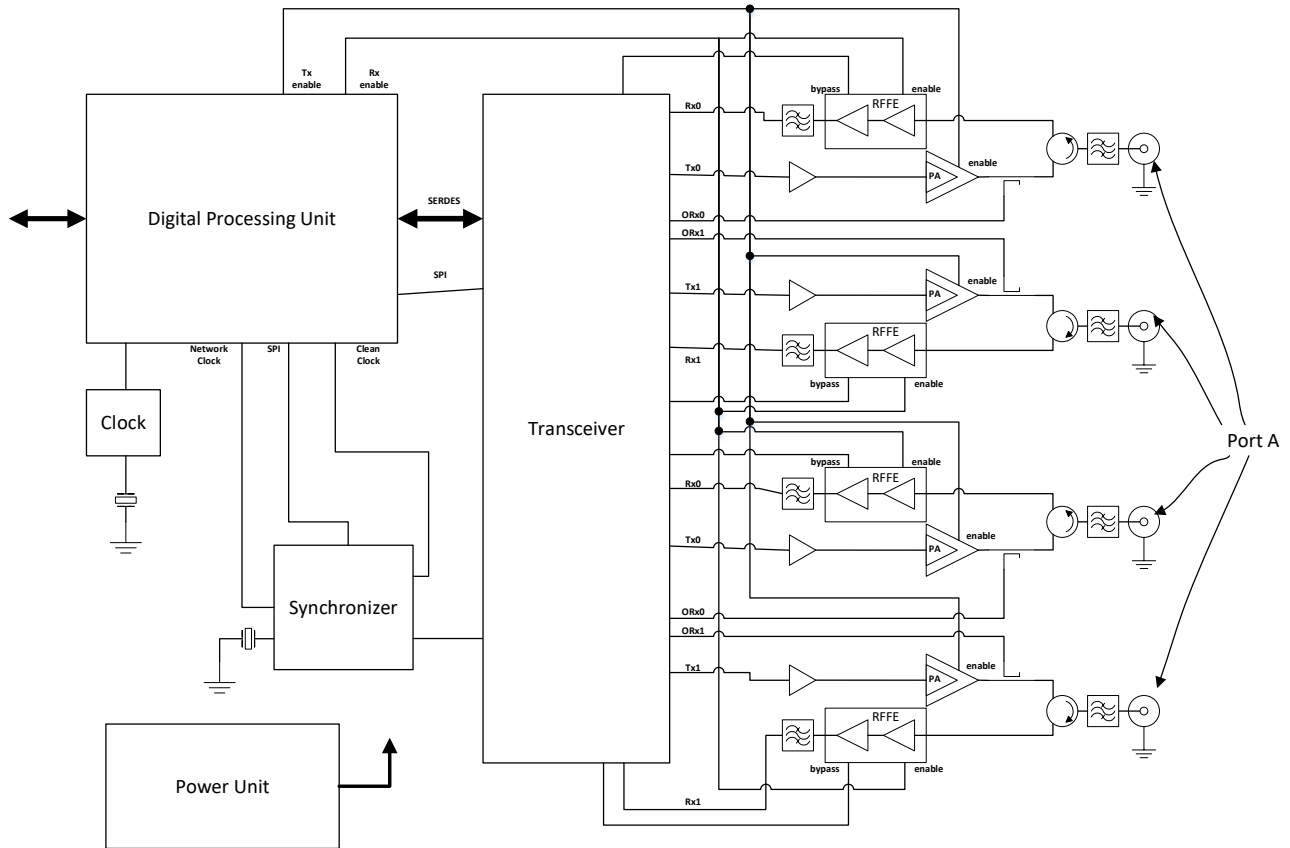


Figure 2.3.2-1 4T4R General Block Diagram with full RF front end

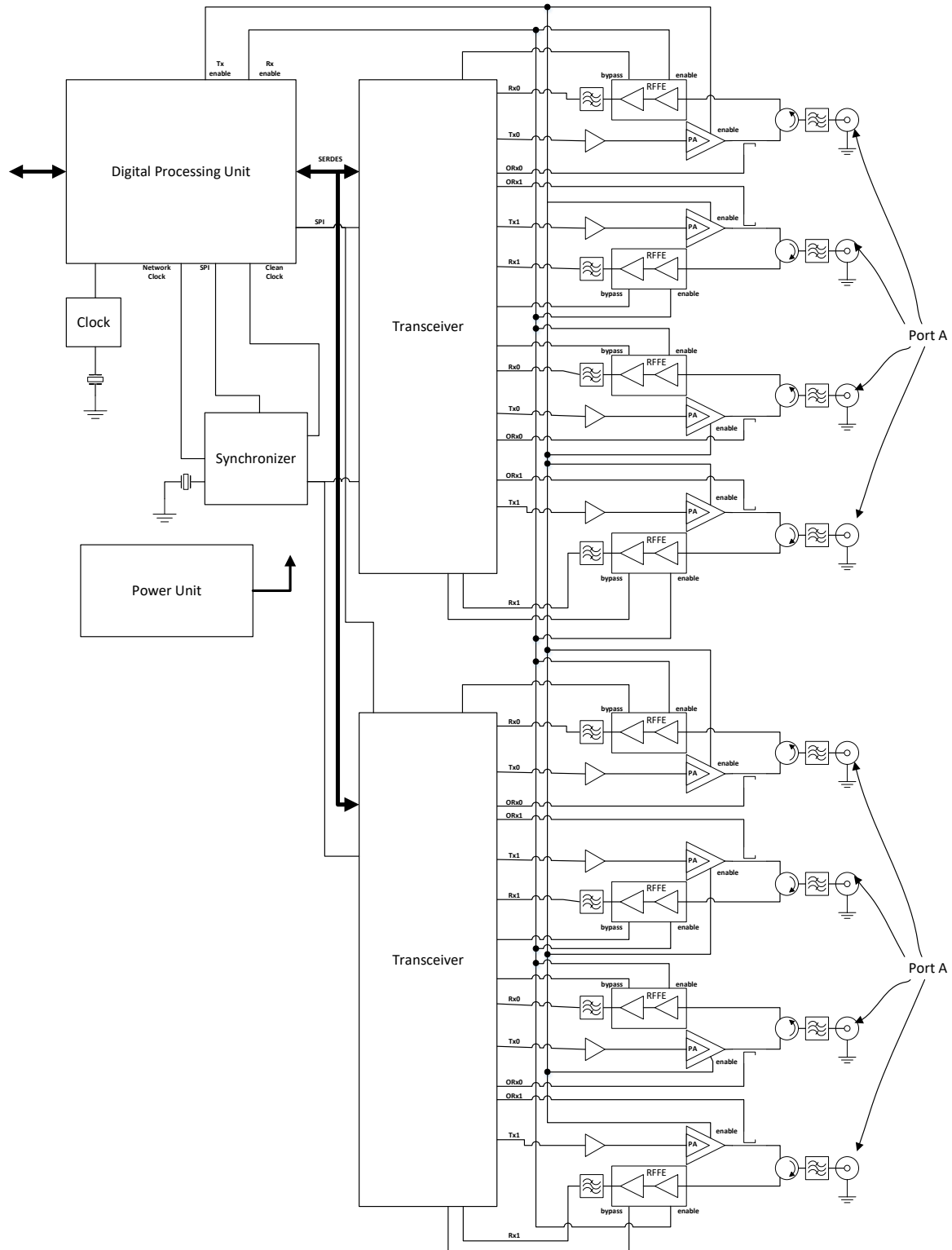


Figure 2.3.2-2 8T8R General Block Diagram with RF front end

Figure 2.3.2-3 shows an example of an high-level block details of an O-RU Hardware implementation. The description of each block from left to right is given below.

An O-RU₇₋₂ communicate with the O-DU₇₋₂ through the O-RAN Fronthaul interface and the control/data packets are processed by the O-RAN Fronthaul unit with the help of the CUS plane processing block. Subsequently the packets are processed by the low-PHY functional blocks such as encoding, scrambling, modulation, layer mapping, precoding, beamforming, and resource element mapping in the downlink. Exactly the reverse operation is done during the uplink process. The output of the low-PHY block is fed to the digital processing unit through packet interface. Here the frequency conversion in digital domain i.e., digital up conversion (DUC) in DL and digital down conversion (DDC) in UL are done. Furthermore, DPD and CFR algorithms are implemented in the digital processing unit to enhance the power amplifier efficiency by reducing the PAPR/ACLR in the RF front end. The output of digital processing unit is fed to the RF processing unit through the digital interface. The digital signals are converted to analog in the DL and vice versa in the UL with the help of DAC and ADC functional blocks in the Transceiver unit, respectively. The signals are further amplified with the help of functional blocks like the PA in DL and LNA in UL both of which are part of the analog RF front end unit. The output of the power amplifier is fed to cavity filter to suppress the unwanted signals in DL and is done before amplification in the case of UL operation i.e., receive operation to improve selectivity. The TDD switch is used to switch the operation between DL and UL, i.e., transmit and receive operations. The output of the cavity filter is fed to antenna unit and then transmitted over the air. The processing units such as FH, Digital, and RF powered from the power unit and clocking and synchronization is provided from the timing/synchronization block. In the case of a beamforming scenarios, antenna calibration is implemented using the antenna calibration processing block. This coordinates with the low-PHY processing unit and calibration switch (Cal switch) and process the signals coming from and going to the antenna calibration port to compensate the amplitude and phase offsets existing in each RF chain. Other implementations are also possible by rearranging the functional blocks of O-RAN FH, Digital processing, and RF processing units.

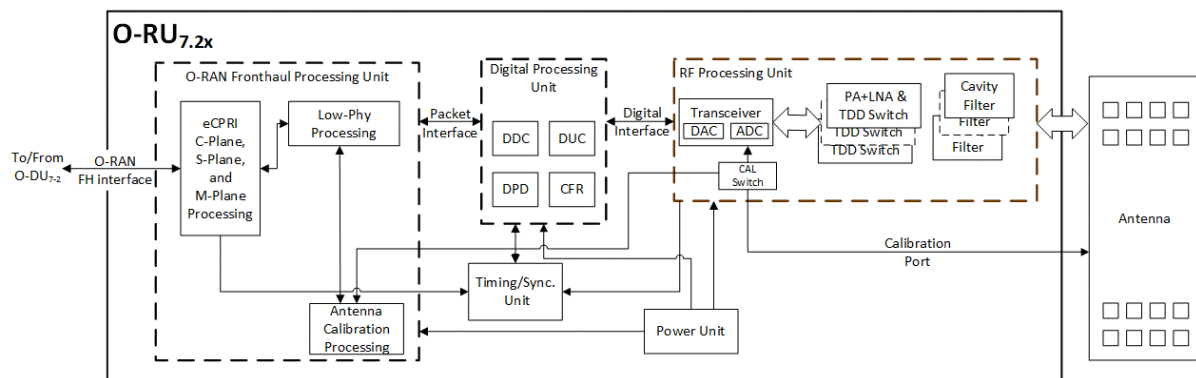


Figure 2.3.2-3 O-RU Architecture Example Implementation 1

2.3.2.1 Digital Processing Unit

The digital processing unit of the O-RU₇₋₂ performs the fronthaul interface, lower L1, Synchronization, RF interface, antenna calibration, digital beamforming and OAM. The following block diagram describes the digital processing unit as per example implementation in

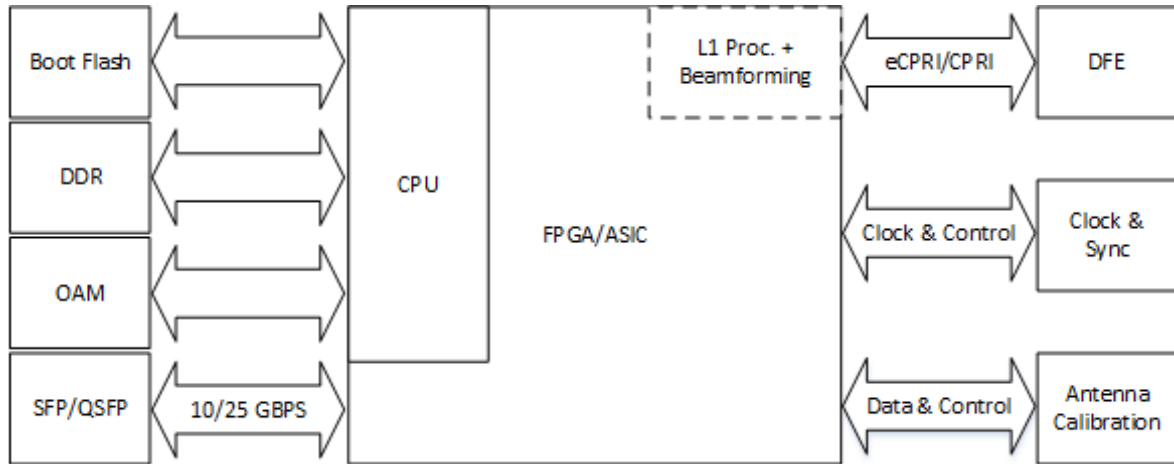


Figure 2.3.2-4 Digital Processing Unit Block Diagram (L1 Processing + optional Beamforming)

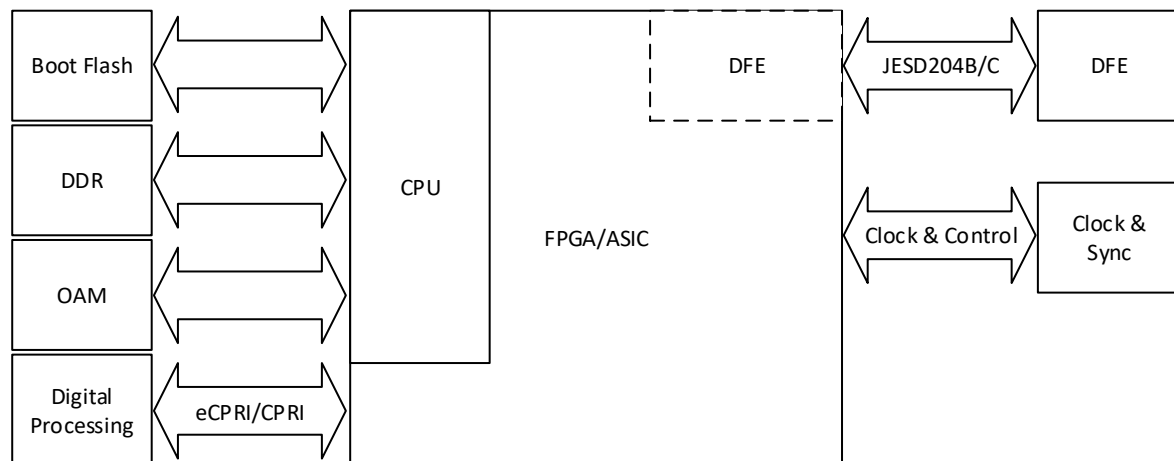


Figure 2.3.2-5 Digital Processing Unit Block Diagram (DFE)

2.3.2.1.1 FPGA/ASIC Solution

This solution for the digital processing unit is based on an FPGA with embedded ARM processor.

2.3.2.1.1.1 FPGA Specifications

Table 2.3.2-2 shows the general interfaces required for an FPGA based design based on Figure 2.3.2-4 and Figure 2.3.2-5.

Table 2.3.2-1 FPGA Interface Specifications

Item Name	Description
eCPRI Port	High speed bi-directional interface transporting data and control information. Total payload will be determined by the antennas/layers served, bandwidth supported, bit precision, subcarrier spacing, etc. It is typically served by one or more QSFP+ or SFP28 connectors depending on the payload.
SERDES	Up to 16 high speed JESD204B/C serial interface lanes between FPGA fabric and transceiver depending on transceiver configuration selected.
Boot Clock	Clock for any uC or fabric functions required immediately at power on
Sysref	JESD204B/C synchronization signal
DevCLK	JESD204B/C device clock
ARM Clock	Clock for the ARM processor
Memory Clock	Memory clock
OAM Port	Control interface used for maintenance
DDR	Interface for external DDR memory

2.3.2.1.1.2 FPGA Design

Table 2.3.2-2 shows the general blocks required for the design. The core elements implemented in this design are required to be sized to support the requirements shown in the OMC-HAR [6] document. Specifically, the DPU should support up to 200 MHz of occupied BW with up to two 100 MHz component carriers. In addition to signal processing, the DPU should support C, U, S and M plane handling. Major processing blocks in the DPU are shown in Table 2.3.2-2.

Table 2.3.2-2 FPGA Functional Blocks

Item Name	Description	Cat A	Cat B
FFT & CP-	FFT and Cyclic Prefix removal for uplink processing	Required	Required
iFFT & CP+	iFFT and Cyclic Prefix insertion for downlink processing	Required	Required
PRACH Front End	PRACH front end processor	Required	Required
Channel DDC	Component Carrier Digital Down Converter	Optional	Optional
Channel DUC	Component Carrier Digital Up Converter	Optional	Optional
JESD204B/C	JEDEC High speed serial interface type B or C framer/de-framer	Required	Required
CFR	Crest Factor Reduction	Optional	Optional
DPD	Digital Pre-Distortion	Optional	Optional
C/U/M Plane	Control, User and management Plane processing	Required	Required
1588v2 Stack	IEEE 1588v2 Synchronization Software Stack	Required	Required
eCPRI	evolved Common Public Radio Interface framer/de-framer	Required	Required
I/Q Compression	I/Q compression/decompression for fronthaul lane rate reduction	Optional	Optional
Beamforming	Create & steer several beams	Optional	Optional
Pre-coding	Pre-coding	Optional	Required
RE Mapping	Resource Element Mapping to Antenna port	Optional	Required

2.3.2.2 RF Processing Unit

The RF Processing Unit of O-RU₇₋₂ includes the power amplifier (PA), low noise amplifier (LNA), Tx/Rx filters and conversion between analog and digital domain. Physical and logical partitioning within the RF processing unit need not occur at any specific boundary.

2.3.2.2.1 Transceiver Reference Design

For the O-RU₇₋₂ the sampling function and frequency conversion function is performed by transceiver. The purpose of the transceiver is to translate the RF signal to digital as early as possible in the signal chain and to reduce the data rate as low as possible to save power. The Transceiver is to convert between high-speed baseband data and a low-level RF for both transmit and receive signal chain. The transceiver may be responsible for orchestration of external gain control elements in both the receive and transmit paths. Other radio functions may need to be controlled by the FPGA/ASIC where they need to be control fast and synchronized with the TDD frame structure. Such elements include the transmit / receive switch, PA and LNA enable.

2.3.2.2.1.1 Hardware Specifications

2.3.2.2.1.1.1 Interface

Table 2.3.2-3 RF Processing Unit Interface Specifications

Item Name	Description
High Speed Data (SERDES Out, SERDES In)	High speed data represents the baseband information being transmitted or received. Depending on the configuration of the ORAN device, various bandwidths may be supported leading to a range of payload

	rates. Options for data include JESD204B and JESD204C. Up to 8 lanes in each direction may be supported although fewer is preferred. Options such as DPD and numeric precision will impact the payload rate. Several options are shown in the following table. All data represents IQ 16-bit (N=16) precision. Some devices support IQ 12 bit (N'=12) which may reduce the required data rates accordingly.
Reference Clock (REF Clock In)	The transceiver should receive a reference for internal clock and LO synthesis needs. This reference clock can function as the JESD204 Device Clock where the interface is by SERDES. The specific clock frequency is determined by the operation mode of the transceiver.
SYSREF	If the transceiver supports SERDES, then it should accept a SYSREF signal from the clock or data source as appropriate. The number and configuration for the SYSREF is dependent on the operating mode of the transceiver.
SYNC\	If the transceiver supports SERDES, then it should also support a SYNCB for each link as appropriate.
Control (SCLK, CSB, SDO, SDIO, etc)	Control of the transceiver is by way of 3 or 4 wire SPI or I2C functioning as a slave. Support for 1.8V control is required and tolerance of 3.3V is preferred. The transceiver may optionally include a separate SPI master for control of peripheral devices as required.
GPIO (GPIO 1 – GPIO #)	The transceiver may optionally include GPIO for controlling peripherals including but not limited to PAs, LNAs and other devices. These GPIOs should at a minimum support 1.8V outputs but the specifics will be determined by the connected devices. The GPIO should also support input from peripheral devices. Input should at a minimum support 1.8V logic with tolerance of 3.3V preferred.
Tx Enable	The transceiver should provide an output to support enabling and disabling the external devices in the transmit chain such as a TxVGA (optional) and PA. This may be part of the GPIO pins.
Rx Enable	The transceiver should provide an output to support enabling and disabling the external devices in the transmit chain such as a RF Front End Module or LNA. This may be part of the GPIO pins.
LNA Bypass	The transceiver should provide an output to support bypassing the LNA appropriately in the condition of a large blocker if so required. This may be part of the GPIO pins.
RF Outputs (Tx1 – Tx4)	RF outputs including the main Tx signal should support 50 ohm or 100 ohms signalling. These outputs can be either single ended or differential.
RF Inputs (Rx1 – Rx4 & ORx 1 – ORx4)	RF inputs including the main Rx and the Observation Rx (ORx) (for DPD) should support 50 ohm or 100 ohms signalling. These inputs can be either single ended or differential. The device should support at least 1 ORx.

Table 2.3.2-4 JESD204B/C Serial Data Rates

Bandwidth	JESD204B	JESD204C	JESD204B	JESD204C
	4T4R		8T8R	
20	4.9152	4.05504	9.8304	8.11008
50	9.8304	8.11008	19.6608	16.22016
100	19.6608	16.22016	39.3216	32.44032
200	39.3216	32.44032	78.6432	68.88064

2.3.2.2.1.1.2 Algorithm

The transceiver is required to provide appropriate algorithms to sustain RF operation including but not limited to Rx AGC, Tx Power control, DPD and CFR.

2.3.2.2.1.1.3 Device Configuration

The transceiver should support either 4T4R or 8T8R. In addition, at least one ORx path should be provided to support DPD functionality.

2.3.2.2.1.1.4 Power Dissipation

Total dissipation of the TRx should be less than 6W for 4T4R.

2.3.2.2.1.1.5 RF Specifications

Table 2.3.2-5 Transceiver RF Specifications

Parameter	Symbol	Min	Typ	Max	Unit	Test Condition/Comment
Transmitters						
Center Frequency		650		6000	MHz	
Transmitter Synthesis Bandwidth				450	MHz	
Transmitter Large Signal Bandwidth				200	MHz	
Transmitter Attenuator Power Control Range		0		32	dB	Signal to noise ratio (SNR) maintained for attenuation between 0 and 20 dB
Transmitter Attenuation Power Control Resolution			0.05		dB	20 MHz LTE/NR at -12 dBFS
Adjacent Channel Leakage Ratio (ACLR)			-66		dB	75 MHz < f ≤ 2800 MHz
In Band Noise Floor			-154.5		dBm/Hz	0 dB attenuation; in band noise falls 1 dB for each dB of attenuation for attenuation between 0 dB and 20 dB
Out of Band Noise Floor			-153		dBm/Hz	600 MHz < f ≤ 3000 MHz

Parameter	Symbol	Min	Typ	Max	Unit	Test Condition/Comment
Maximum Output Power	OIP3	6			dBm	0 dB attenuation; 3 × bandwidth/2 offset
Third Order Output Intermodulation Intercept Point		27			dBm	600 MHz < f ≤ 3000 MHz
Error Vector Magnitude (3GPP Test Signals)	EVM					
1900 MHz LO		0.6			%	50 kHz RF PLL loop bandwidth
3800 MHz LO		0.53			%	300 kHz RF PLL loop bandwidth
Observation Receivers Center Frequency	ORx	450	6000		MHz	IIP3 improves dB for dB for the first 18 dB of gain attenuation; QEC performance optimized for 0 dB to 6 dB of attenuation only. For attenuation steps from 0 dB to 6 dB
Gain Range			30		dB	
Analog Gain Step	P _{HIGH}	0.5			dB	
Receiver Bandwidth				450	MHz	0 dB attenuation; increases dB for dB with attenuation. 450 MHz Integration bandwidth 491.52 MHz Integration bandwidth
Maximum Usable Input Level		-11			dBm	
Integrated Noise	IIP3	-58.7			dBFS	Maximum observation receiver gain; test condition: P _{HIGH} -11 dB/tone
		-57.5			dBFS	
Third Order Input Intermodulation Intercept Point Narrow Band						
1900 MHz	IM3	15			dBm	IM3 products > 130 MHz at baseband; test condition: P _{HIGH} - 11 dB/tone; 491.52 MSPS
2600 MHz		16.5			dBm	
3800 MHz		18			dBm	
Wide Band	IM3					600 MHz < f ≤ 3000 MHz
1900 MHz		13			dBm	
2600 MHz		11			dBm	
3800 MHz		13			dBm	
Third Order Intermodulation Product		-70			dBc	

Parameter	Symbol	Min	Typ	Max	Unit	Test Condition/Comment
Spurious Free Dynamic Range	SFDR		64		dB	Non IMx related spurs, does not include HDx; ($P_{\text{HIGH}} - 11$) dB input signal
Harmonic Distortion Second Order Harmonic Distortion Product	HD2		-80		dBc	($P_{\text{HIGH}} - 11$) dB input signal In band HD falls within ± 100 MHz
			-73		dBc	Out of band HD falls within ± 225 MHz
Third Order Harmonic Distortion Product	HD3		-70		dBc	In band HD falls within ± 100 MHz
			-65		dBc	Out of band HD falls within ± 225 MHz
Receivers Center Frequency Analog Gain Step			30	6000	MHz dB	Attenuator steps from 0 dB to 6 dB
			1		dB	Attenuator steps from 6 dB to 30 dB
Receiver bandwidth Maximum Usable Input Level	P_{HIGH}			200	MHz	0 dB attenuation; increase dB for dB with attenuation; continuous wave = 1800 MHz; corresponds to -1 dBFS at ADC
Noise Figure	NF		-11		dBm	75 MHz $< f \leq 3000$ MHz
Input Third Order Intercept Point Difference Product	IIP3 IIP3, d		12		dB	0 dB attenuation at receiver port 600 MHz $< f \leq 3000$ MHz
2600 MHz (Wideband)			17		dBm	Two ($P_{\text{HIGH}} - 9$) dB tones near band edge
2600 MHz (Midband)			21		dBm	
Sum Product	IIP3, s					Two ($P_{\text{HIGH}} - 9$) dB tones, at bandwidth/6 offset from the LO
2600 MHz (Wideband) HD3	HD3		20		dBm	
			-66		dBc	($P_{\text{HIGH}} - 6$) dB continuous wave tone at bandwidth/6 offset from the LO 600 MHz $< f \leq 4800$ MHz

2.3.2.2.1.2 Hardware Design

For the O-RU₇₋₂ the sampling function and frequency conversion function can be performed by the transceiver. The transceiver integrates the ADC, DAC, LO, down converter, up converter and related functions. The block diagram of transceiver design is shown in Figure 2.3.2-6.

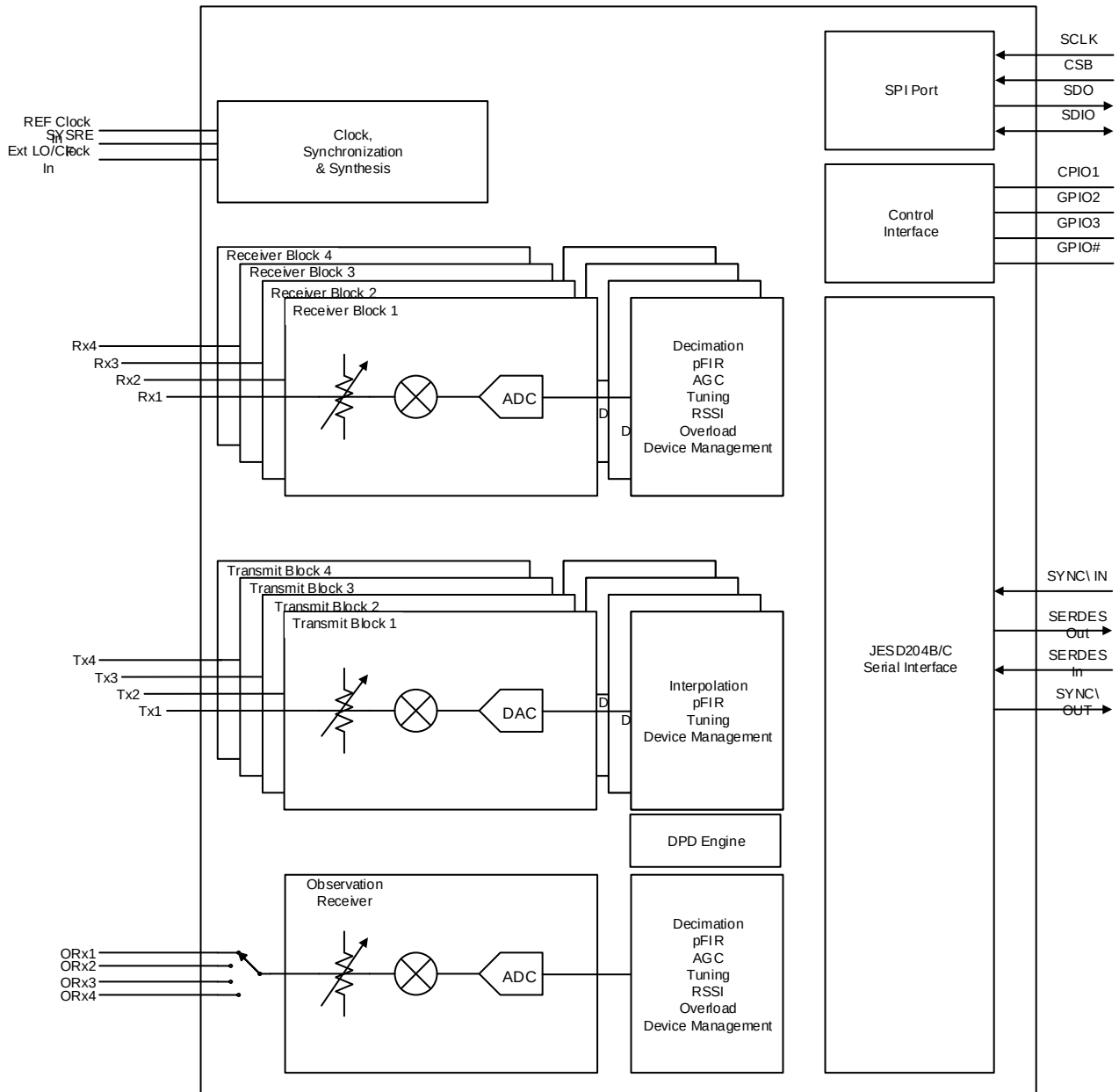


Figure 2.3.2-6 Transceiver with Integrated CFR and DPD

2.3.2.2.2 Power Amplifier (PA) Reference Design

2.3.2.2.2.1 Hardware Specifications

2.3.2.2.2.1.1 Interface

Table 2.3.2-6 Power Amplifier Interface Specifications

Item Name	Description
RF Enable	The enable input should be compatible with 1.8V logic and tolerate 3.3V as required. A logic high enables the PA. A logic low disables the device and places it in a minimum dissipation mode.
RF Output	RF outputs support 50-ohm single ended to properly interface to a directional coupler, isolator, switch or antenna.
RF Input	RF inputs should support 50-ohm, single ended match to the transceiver output or preamp.

2.3.2.2.2.1.2 Power Specifications

RF output power required is nominally 10 W (40 dBm). DC power is driven by efficiency of the PA and should not be more than 27W and the efficiency should be greater than 37%.

2.3.2.2.2.1.3 RF Specifications

The PA and driver should have enough gain to boost the transceiver output to the rated power level. The output power should be at least 10W (40 dBm) including the loss of the circulator and antenna filter. The ACLR with DPD and CFR should be better than 47dBc according to the related 3GPP test models.

Table 2.3.2-7 Power Amplifier RF Specifications

Item Name	n 77, 48	n 41
Gain	>60 dB including driver	To be provided in next version
P3dB	51 dBm	
Input Return Loss	<-15 dB	
Output Return Loss	<-15 dB	
Switching Speed	<1uS	
HD2	>28 dBc	
HD3	>30 dBc	

2.3.2.2.2.2 Hardware Design

RFin is the input of the PA, RFout is the output of the PA. Vdd and Vbias is the power input of the PA. PA_{enable} is the control pin to disable or enable the PA. The input and output should be matched to 50 ohms to optimize return loss as much as possible. Proper decoupling shall be provided to minimize impact of supply ripple on the RF output.

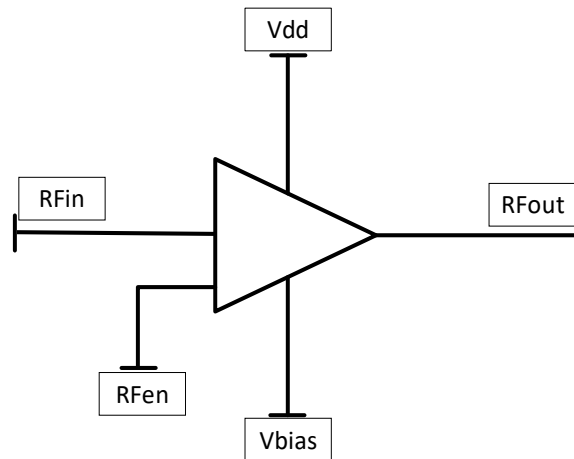


Figure 2.3.2-7 Power Amplifier Reference Design

2.3.2.2.3 Low Noise Amplifier (LNA) Reference Design

2.3.2.2.3.1.1 Interface

Table 2.3.2-8 LNA Interfaces

Item Name	Description
Enable	The enable input should be compatible with 1.8V logic and tolerate 3.3V as required. A logic high enables the LNA. A logic low disables the device and places it in a minimum dissipation mode.
LNA Bypass	The LNA Bypass skips the second stage LNA effectively reducing the overall gain. This would be selected when the input signal compromises the linearity of the amplifier.
RF Out	RF outputs support 50-ohm single ended to properly interface to the RF filtering and Rx input port of transceiver.
TR Switch	The TR switch is used to shunt any reflected RF power that reaches the LNA to a shunt load during the transmit period.
RF Input	RF inputs should support 50-ohm, single ended match to circulator.
Termination	This RF output port shunts any reflected transmit power to an appropriate load during the Transmit period to protect the input of the LNA.

2.3.2.2.3.1.2 Power Specifications

DC power per channel should not exceed 0.5W.

2.3.2.2.3.1.3 RF Performance Specifications

Table 2.3.2-9 LNA RF Specifications

Item Name	n 77, 48	n 41
Gain (High/Low)	32/16 dB	To be provided in next version
NF (High/Low)	1.5/1.5 dB	
OIP3 (High/Low)	31/31 dBm	
OP1dB (High/Low)	18/13 dBm	
TR Switch Time	<1uS	

2.3.2.2.3.2 Hardware Design

Figure 2.3.2-8 shows the configuration for a dual-stage LNA. The RF inputs and outputs should be properly matched to optimize the noise performance and flatness of the amplifier. The input switch is used to forward any reflected transmit energy to an external load during the transmit period to protect the LNA from damage. The DC supply voltage, VCC, should be properly filtered to prevent power supply noise from entering the LNA and reducing performance.

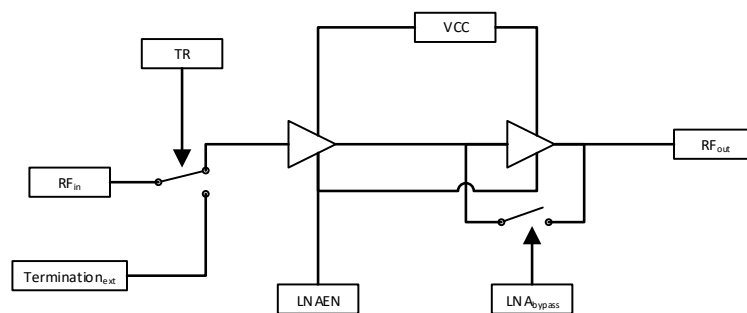


Figure 2.3.2-8 LNA Reference Design

2.3.2.2.4 Circulator Reference Design

For TDD use, the Tx and Rx links work in time duplex. To properly couple the Tx energy to the antenna and the Rx signal to the LNA input, a circulator is used to perform this function given the potentially high RF power being processed through this device. The circulator can be configured as a clockwise or counterclockwise device depending on the layout details.

2.3.2.2.4.1 Hardware Specifications

2.3.2.2.4.1.1 Interface

The circulator includes 3 ports that can function as in input or an output. If the device is labelled clockwise, then for each input, its nominal output is the next pin rotating right around the device. If it is counterclockwise, the nominal output is the next pin rotating to the left around the device.

2.3.2.2.4.1.2 RF Specifications

Table 2.3.2-10 Circulator Specifications

Item Name	n 77, 48	n 41
Rotation	CW or CCW	To be provided in next version
Average Power	Up to 46 dBm	
Peak Power	52 dBm	
Impedance	50 ohms	
Insertion Loss	-0.3 dB	
Isolation	20 dB	
Return Loss	22 dB	

2.3.2.2.4.2 Hardware Design

The example on the left in Figure 2.3.2-9 is a clockwise oriented device. The RF input enters pin 1 and exits pin 2. Pin 2 can also be used as an input and its output is pin 3. Finally pin 3 can also function as an input with pin 1 being the output. The device on the right is counterclockwise oriented and the signal flow is the in the opposite direction.

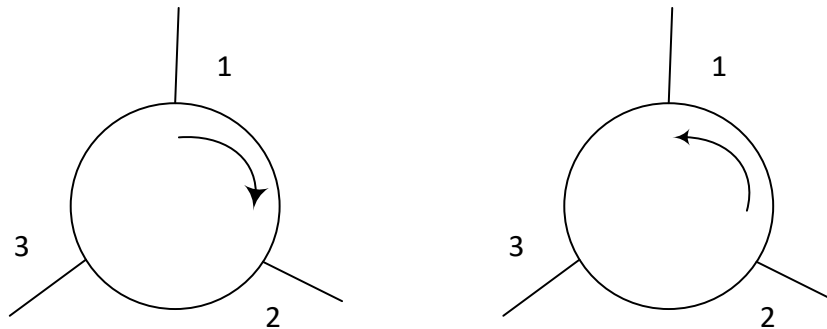


Figure 2.3.2-9 Circulator Reference Design

2.3.2.2.5 Antenna / Phased Array Reference Design

The antenna is a transducer, and it is used to convert electrical signal to the Electromagnetic waves during the transmit mode and vice versa in the reception mode. radiate the TX power and to receive the RX signal. Based on the radiation pattern characteristics of the antenna, it is broadly classified as omni-directional and directional antennas and examples of such antennas are Two options are listed below for reference.

2.3.2.2.5.1 Hardware Specifications for Omnidirectional Antenna

The following table shows the typical electrical omnidirectional antenna specifications for the O-RU₇₋₂.

Table 2.3.2-11 Omnidirectional Antenna Specifications

Frequency band	n 77	n 48	n 41
Antenna Type	Omnidirectional	Omnidirectional	To be provided in next version
VSWR (max)	1.5:1	1.5:1	
Power capacity	≥10 W	≥10 W	
Gain (max)	5.5 dBi	5.5 dBi	
Beam width (vertical)	35°	35°	
Polarization	±45°	±45°	

2.3.2.2.5.1.1 Hardware Design for Omnidirectional Antenna

One possible choice of the omnidirectional antenna is a canister type with multiple ports including the 4 ports for the 4T4R radio. Conceptual design is shown in the following figure.


Figure 2.3.2-10 Canister Style Omni Antenna

2.3.2.2.5.2 Hardware Specifications for Directional Antenna

The following table shows the typical electrical directional antenna specifications for the O-RU₇₋₂.

Table 2.3.2-12 Directional Antenna Specifications

Frequency band	n 77	n 48	n 41
Antenna Type	Directional	Directional	To be provided in next version
Antenna with Embedded RET & AISG[10] Compliant	Antenna with Embedded RET & AISG[10] Compliant	Antenna with Embedded RET & AISG[10] Compliant	Antenna with Embedded RET & AISG[10] Compliant
VSWR (max)	1.5:1	1.5:1	1.5:1
Power capacity	≥10 W	≥10 W	≥10 W
Gain (max)	16 dBi	16 dBi	16 dBi
Beam width (horizontal)	65°	65°	65°
Beam width (vertical)	11°	11°	11°
Polarization	±45°	±45°	±45°
Variable Tilt	0-10 °	0-10 °	0-10 °
Tilt Accuracy	±1 °	±1 °	±1 °
3rd Order PIM, @2x43 dBm (dBc)	≤-153	≤-153	≤-150
Isolation (RF Ports) (dB)	≥25	≥25	≥20

2.3.2.2.5.2.1 Hardware Design for Directional Antenna

One possible choice of directional antennas are the radios mounted on the back of the antenna with 4 ports for the 4T4R radio on the back of the antenna just below each radio. Conceptual design is shown in the following figure.

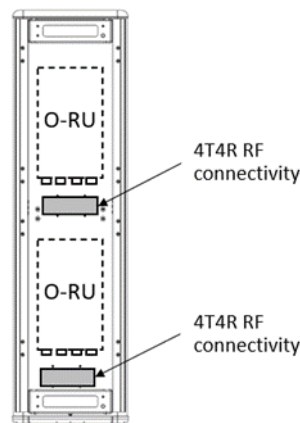


Figure 2.3.2-11 Mounting arrangement scenario 1 of 4T4R radios on the back of Directional Antenna

Another possible choice of directional antennas are the radios located away from the antenna with ports for each of the 4T4R radios on the bottom cap of the antenna. Conceptual design is shown in the following figure.

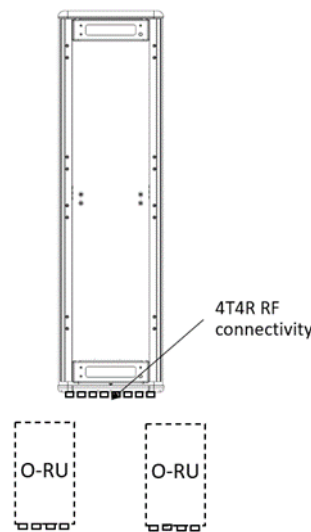


Figure 2.3.2-12 Mounting arrangement scenario 2 of 4T4R radios remotely mounted with Directional Antenna

2.3.2.3 Internal Interfaces

The following sections describe potentially useful optional interfaces within the white box. These optional interfaces are not required and may be used at the discrepancy of the manufacturer. This section is provided strictly as information only to better clarify the functionality of these interfaces.

2.3.2.3.1 Optional JESD204 Interfacing

O-RU solutions requiring chip to chip interfacing of high-speed time domain data may benefit from utilization of JEDEC defined standard JESD204 [16]. This interface is specified for chip-to-chip interfacing that provides fast, low latency and time aligned data. The benefit of such an interface is to enable an open standard that allows for devices from multiple vendors to seamlessly interface. From a logistical perspective JESD204 allows multiple devices to communicate over the same link in a multiplex fashion while preserving time alignment and providing deterministic latency. This makes JESD204 a good candidate interface for carrying the signals for multiple antennas, multiple component carriers including I and Q data. This allows for relative simplicity requiring only a few differential pairs along with a ‘device clock’ and ‘sysref’ signals to carry and synchronize all time aligned signals.

Figure 2.3.2-13 provides an example of a typical configuration for a JESD204 configuration. The lines labeled ‘JESD for UL x2’ and ‘JESD for DL x2’ are the data bearing connections and consist of differential pairs. In this example, there are 2 differential pairs for the UL data (receiver path) and 2 differential pairs for the DL data (transmit path). These connections provide data using an embedded clock and synchronization with a transport layer coding used to ensure sufficient transitions that the embedded clock can be recovered on the receiving end.

The second signal of is labeled ‘Device Clock’. The device clock provides the frequency reference to devices in the configuration. Within the scope of JESD204, this signal provides the reference for the interface. In practice, the Device Clock also provide the frequency reference to the individual devices and is usually a

multiple of 30.72 MHz. JESD204 connected devices would use this reference for both the interface as well as other frequency services within the device such as sample clocks and other PLLs.

The final signal is the ‘Sysref’ which enables deterministic latency in the system. Each interconnected device uses this signal in conjunction with ‘Device Clock’ to determine exactly when the sample edge should occur. There are different ways to configure and use ‘Sysref’ to achieve different goal. For more details on configuration, see [16].

JESD204 is an evolving standard. Although it has been around for a while, as process technology evolves and data needs grow, JESD has evolved to keep pace with throughput needs. The initial variation supported up to 3.125 Gbps. JESD204A brought speeds up to 6.25 Gbps. JESD204B brought throughput to 12.5 Gbps. Most recently JESD204C offers throughputs up to 32.45 Gbps. Each variety offers at least one option for transport layer coding like 8B/10B as well as a range of other features including sub-channel coding which allows control channels to be passed along with the primary data. Currently most devices support JESD204B and JESD204C. As of this writing, JEDEC is underway defining the next phase of JESD204x which will further increase data rates as demanded by increasing bandwidth in wireless and other application.

Table 2.3.2-13 JESD204 Abridged Lexicon

Parameter	Description
M	Number of converters/converter device (link)
NP (or N')	Total number of bits per sample
S	Samples transmitted/single converter/frame cycle
F	Octets/Frame
K	Number of frames per multi-frame
SC	Sample Clock Rate
L	Number of lanes/converter device (link)

To better comprehend the JESD204 standard, understanding the lexicon is important. There are many parameters that can be used, but those in Table 2.3.2-13 provide the core definitions. Most of these are basic parameters related to common data converter characteristics and include the number of converters, number of bits and sample rate. The remaining parameters are related to the transport layer and physical implementation of the interface.

Combined, the parameters in this lexicon help define the data throughput, number of lanes required as well as the framing of data onto the interfaces. Starting with the payload, it is determined by computing the serial data rate and adding the transport layer (ie 8B/10B, 64B/66B etc) to that as defined in the following equation:

$$M \times NP \times SC \times \text{Transport Layer Coding} = \text{Payload Rate}$$

This equation provides the total payload to be delivered between devices. This data is delivered over one or more lanes. If they payload exceeds the capacity of one lane, additional lanes are added to ensure that the throughput per lane matches the capability of the standard. As highlighted in preceding paragraphs, currently JESD204B supports up to 12.5 Gbps and JESD204C supports up to 32.45 Gbps but future evolutions will include higher lane capacity. Where the Payload Rate exceeds an individual lanes capacity, the total Payload Rate is divided by the lane capacity, currently 12.5 Gbps or 32.45 Gbps, to determine the approximate number of lanes required. The actual number is the next higher integer to ensure that the lane rate is less than the

maximum specified or desired. As the standard continues to evolve, this same math will be used to determine the payload and number of lanes, L.

$$L = \text{Celing} \left(\frac{\text{Payload Rate}}{\text{Max Lane Rate}} \right)$$

The final important term is F, the number of octets per frame. In JESD204, an octet represents an encoded 8-bit word. Fundamentally, F defines the frame structure given the number of lanes and the specific payload to deliver. F is defined by the following equation and represents the number of octets per frame:

$$F = \frac{M \times X \times NP}{8 \times L}$$

Figure 2.3.2-13 provides an example 4T4R configuration supporting a typical 100 MHz waveform. The low PHY device processes complex (I & Q data) for each datapath and thus there are a total of 8 data streams that must be moved between the digital chip and the data conversion. The bit precision of this data is sufficient to meet the noise requirements of the solution and could be 12 or more bits. For this example, 16-bits will be assumed. To support a 100 MHz bandwidth, a complex data rate of 122.88 MSPS will be used as indicated by Table B.5.2-2 of [3].

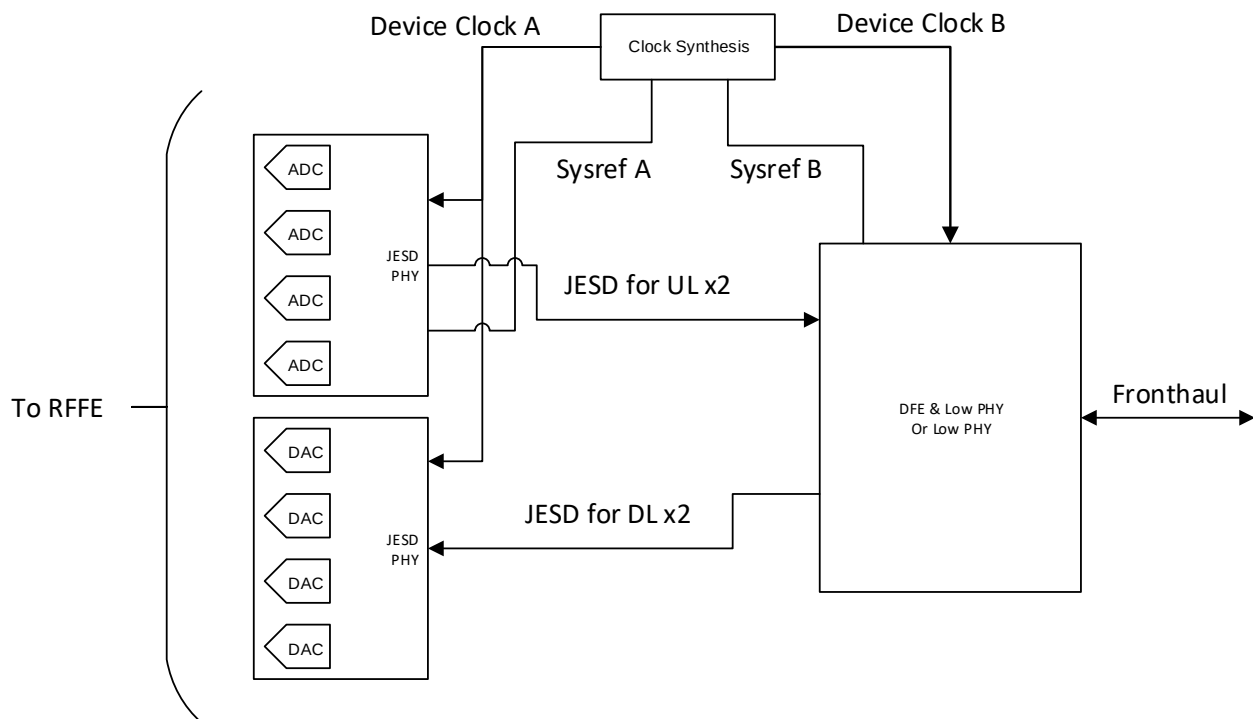


Figure 2.3.2-13 Example 4T4R JESD204 Configuration

Using this information and the equations above and applying 8B10B transport layer coding, the total payload, number of lanes and octets per frame can be computed. Since this is a 4T4R application and each path is

represented by a complex stream, ‘M’ equals 8. ‘NP’ is 16 bits. ‘SC’ equals 122.88 MSPS. Total data payload can be computed using the following equation:

$$Payload = 8 \times 16 \times 122.88^6 \times \frac{10}{8} = 19.6608 Gbps$$

Assuming the interface is based on JESD204B, then the maximum data rate for a single lane would be 12.5 Gbps. Therefore, we need to divide the data onto two physical lanes of 9.8304 Gbps each. Hence in this example, ‘L’ equals 2.

$$L = Ceiling\left(\frac{19.6608 Gbps}{12.5 Gbps}\right) = 2$$

Finally, ‘F’ the number of Octets/Frame is computed to be 8.

$$F = \frac{8 \times 1 \times 16}{8 \times 2} = 8$$

Using this information, the mapping of data can be determined by using online tools such as [17]. These tools describe how the converter data is mapped onto each lane by octet as shown in Figure 2.3.2-14. In this example on lane 0, the first octet conveyed are the most significant bits of converter 0. This is followed by the LSBs in the second octet. This is followed by converter 1, 2 and 3 on lane 0. In parallel, lane 1 provides the data for converters 5 through 7 with the most significant octet leading. In this figure, M#SO represents the logical converter where # identifies the specific converter.

4 (L = 2, M = 8, F = 8, S = 1, NP = 16)								
Frame 0								
Lanes	Octet 0	Octet 1	Octet 2	Octet 3	Octet 4	Octet 5	Octet 6	Octet 7
0	M0S0[15:8]	M0S0[7:0]	M1S0[15:8]	M1S0[7:0]	M2S0[15:8]	M2S0[7:0]	M3S0[15:8]	M3S0[7:0]
1	M4S0[15:8]	M4S0[7:0]	M5S0[15:8]	M5S0[7:0]	M6S0[15:8]	M6S0[7:0]	M7S0[15:8]	M7S0[7:0]

Figure 2.3.2-14 4T4R Example Data Mapping

More complicated examples are also supported as shown in Figure 2.3.2-15 where an 8T8R is described. This configuration supports a 200 MHz iBW and thus requires a complex data rate of 245.76 MSPS. In this case, there are 8 transmit and 8 receive paths. Each of these requires a complex path to the low PHY device.

Therefore ‘M’ equals 16. Each of these are represented by 16 bits. In this example, JESD204C is assumed and thus 64B/66B coding is used. Thus, the total throughput becomes:

$$16 \times 16 \times 245.76^6 \times \frac{66}{64} = 64.88064 Gbps$$

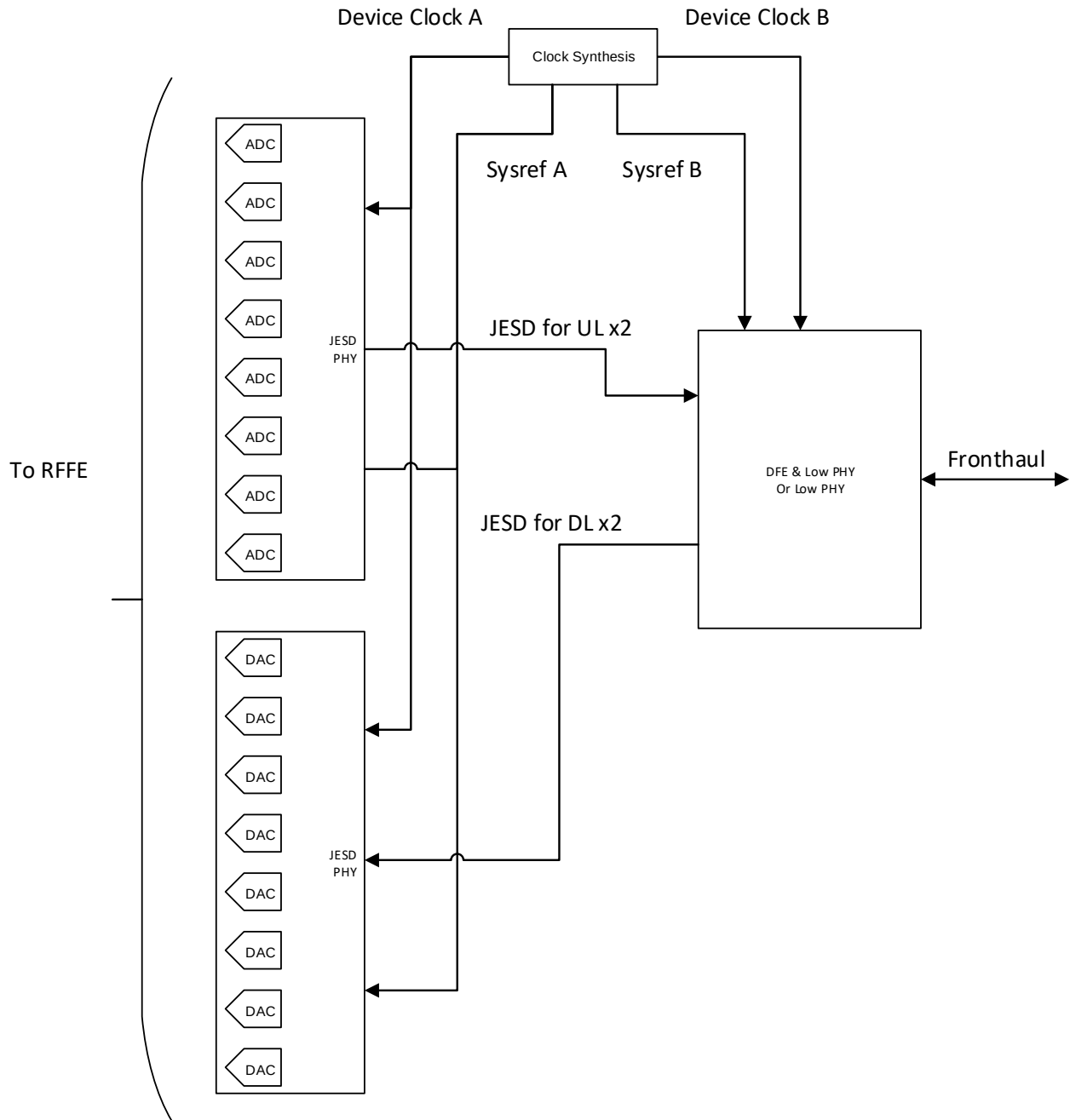


Figure 2.3.2-15 Example 8T8R JESD204 Configuration

JESD204C supports up to 32.45 Gbps and therefore 2 lanes will be required as shown below. In this example, 2 UL and 2 DL JESD lanes connect between the two devices.

$$L = \text{Ceiling} \left(\frac{64.88064 Gbps}{32.45 Gbps} \right) = 2$$

Using the equation for ‘F’ determines the octets per frame to be 16 in this example. The frame structure as defined by JESD204 is shown in Figure 2.3.2-16 As with the 4T4R example, the frame structure is little endian with lane 0 beginning with converter 0 and lane 1 beginning with converter 8. Complete details on JESD204 can be found [16].

$$F = \frac{16 \times 1 \times 16}{8 \times 2} = 16$$

8 (L=2, M=16, F=16, S=1, NP=16)																
Frame 0																
Lanes	Octet 0	Octet 1	Octet 2	Octet 3	Octet 4	Octet 5	Octet 6	Octet 7	Octet 8	Octet 9	Octet 10	Octet 11	Octet 12	Octet 13	Octet 14	Octet 15
0	M0S0[15:8]	M0S0[7:0]	M1S0[15:8]	M1S0[7:0]	M2S0[15:8]	M2S0[7:0]	M3S0[15:8]	M3S0[7:0]	M4S0[15:8]	M4S0[7:0]	M5S0[15:8]	M5S0[7:0]	M6S0[15:8]	M6S0[7:0]	M7S0[15:8]	M7S0[7:0]
1	M8S0[15:8]	M8S0[7:0]	M9S0[15:8]	M9S0[7:0]	M10S0[15:8]	M10S0[7:0]	M11S0[15:8]	M11S0[7:0]	M12S0[15:8]	M12S0[7:0]	M13S0[15:8]	M13S0[7:0]	M14S0[15:8]	M14S0[7:0]	M15S0[15:8]	M15S0[7:0]

Figure 2.3.2-16 8T8R Example Data Mapping

2.3.2.3.2 Optional Die to Die Interfacing

The existing specification O-RAN.WG7.OMC-HRD-Opt7-2-v03.00 has several references to SERDES links, either in text or diagrams. The SERDES link is very appropriate for deployments where RF and baseband chips are separated by 10s of cm on a PCB, as it provides low wire count with decent power consumption (pJ/bit). For multi-chip modules, however, where die are separated by 10 mm or less, die-to-die (D2D) parallel interfaces such as Advanced Interface Bus (AIB) [18], Bunch of Wires (BoW) [19] or Universal Chiplet Interconnect Express (UCIe) [20] offer a lower power, lower complexity option, at the cost of more I/O.

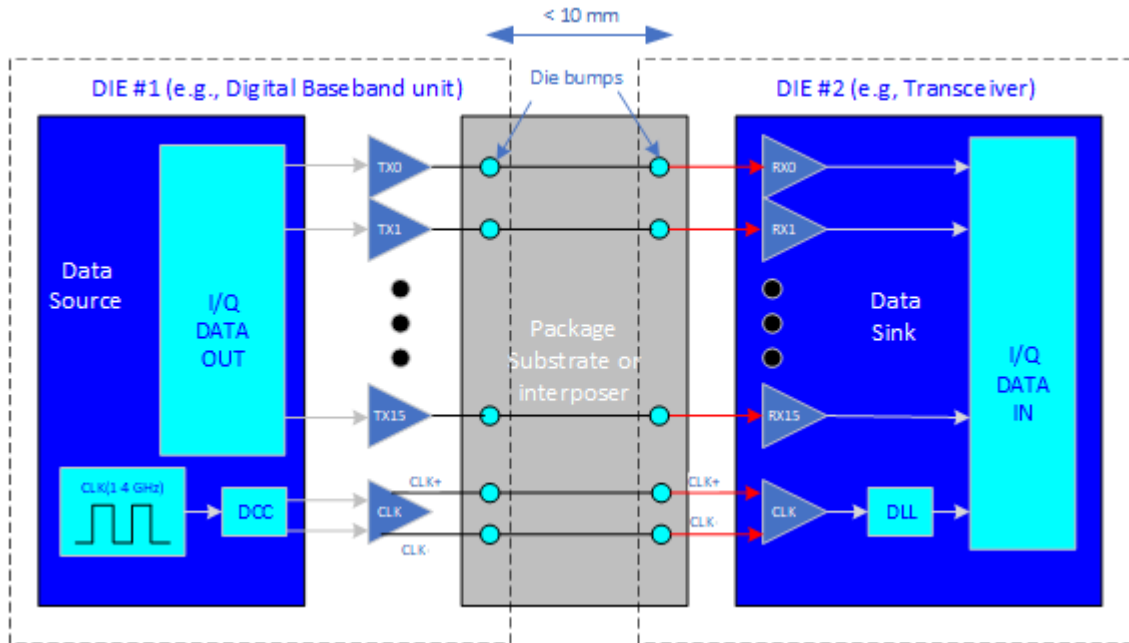


Figure 2.3.27-17 Basic die-to-die parallel architecture

The simplicity of the die-to-die parallel architecture is illustrated in the figure above which depicts a 16-bit unidirectional “data unit” with single-ended drivers and receivers, plus a differential forwarded clock. The salient features of die-to-die interfaces are:

1. Uni-directional throughput rates in the 100-1000 Gbps/mm depending on packaging technology, clock rate, channel length, and use of source/load terminations. ¹
2. Power consumption in the .5-1.0 pJ/bit range.
3. Scaleable, via the addition of more data units consisting of 16-20 I/O each, encompassing more silicon beachfront or bump rows.
4. Simple PHY architecture not requiring any complex pre-emphasis or equalization...usually only Duty Cycle Correction (DCC) and a Delay Lock Loop (DLL).
5. Reliable link such that FEC is typically not required.

The diagram above shows an implementation of D2D in a Multi-chip module including the Digital Baseband unit and the Transceiver in one package. Referring to Figure 2.3.2-2, the Digital Processing unit in the figure corresponds to the Digital Baseband unit in the figure above. This figure shows DL I/Q data flowing to the transceiver, but there would be a corresponding path in the other direction for UL I/Q data flowing from Transceiver to the Digital Baseband unit.

In Figure 2.3.27-17, the clock source is shown on the left of the data transmitting unit. Since DDR mode may be used for max throughput, and the clock is typically in the 1-4 GHz range, an accurate duty cycle is required. The Duty Cycle Correction (DCC) block would be employed to ensure that a duty cycle close to 50% is maintained.

On the RX side, a Delay Lock Loop (DLL) may be used to ensure that the clock strobes data into the data register at approximately the middle of the data bits. If so, a calibration or initialization procedure would determine the clock phase shift required to center the strobe.

The packaging technology and bump pitch used for each die greatly influences the final throughput expressed as Gbps/mm. Standard chip scale bump pitches of 100 um or greater, used with organic substrate chip scale packaging, allow for simplified D2D interconnect but will not give the maximum throughput. The use of micro-bumps and specialized interposer technology, or even EMIB, will provide for throughput approaching 1 Tbps/mm and lower power.

With the AIB, UCIe, and BoW D2D interfaces mentioned initially, a standardized bump map for each data unit is either required, or strongly recommended. (At a minimum, there needs to be a wiring order so that packages with different bump pitches can still be used and data lines routed through the RDL without criss-crossing or excess routing skew.) The # of routing layers in the packaging interconnect determines how many data units (with their associated physical bump patterns) can be adjoined (inwards towards the die), in order to further increase throughput for the same silicon beachfront.

Initial D2D deployments used un-terminated drivers and receivers, which was sufficient for clock rates in the low GHz range and channel lengths of a few mm. To achieve longer channel lengths and/or higher base clock rates, terminations may be employed at the source driver, the receiver, or both.

¹ Here the linear mm dimension refers to the silicon edge beachfront dedicated to the parallel I/O

2.3.3 Synchronization and Timing

The clock and synchronization are determined by the PTP and SyncE signals transported by ethernet L2 via the fronthaul in conjunction with clock recovery in the hardware. Together these provide the local time of day and frequency reference for the O-RU₇₋₂.

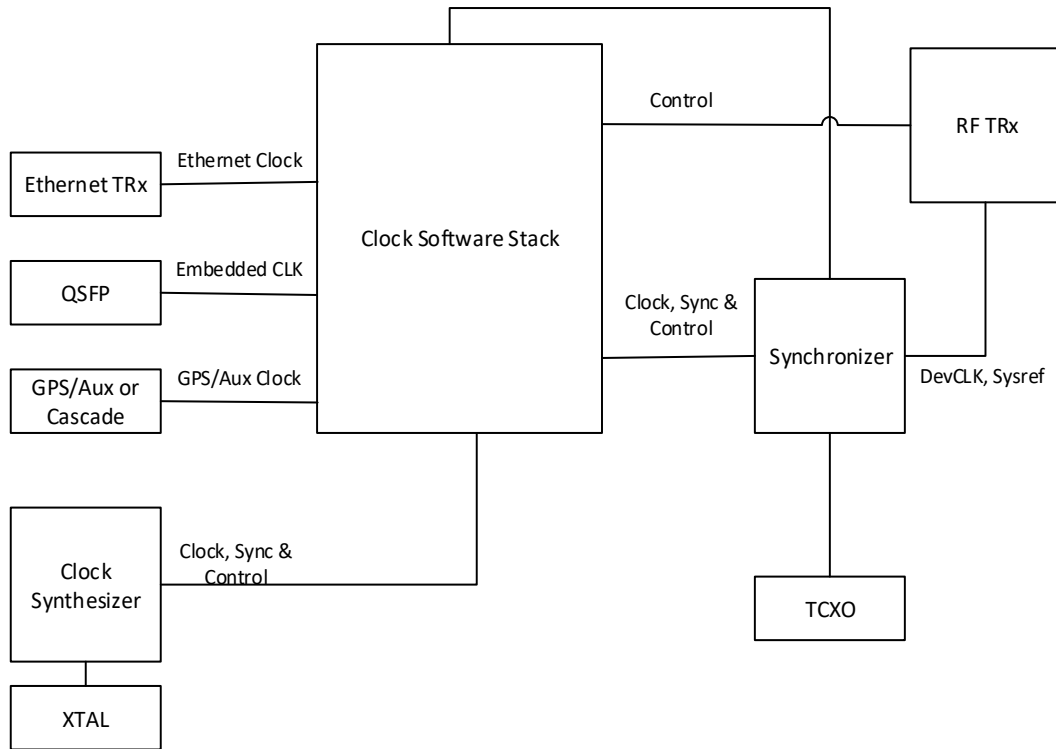


Figure 2.3.3-1 Clock and Synchronization Functional Block Diagram

2.3.3.1 Hardware Specifications

2.3.3.1.1 Interface

Table 2.3.3-1 Synchronization and Timing Interface Specifications

Item Name	Description
Clock Inputs	Depending on the configuration, the following clock inputs may exist: QFSP, Ethernet, GPS, Cascaded unit clocks, Synchronized Clock.
Clock Outputs	Depending on functional requirements, the following clocks may be delivered: QFSP, Ethernet, DDR, CPU, RF Transceiver clock, Synchronizer reference.
Clock Control	SPI or I2C to control the clock synthesizer and synchronizer.
Clock Sync	Logic signal to phase align and time align the synchronizer with master.
Crystal Reference	One or more reference oscillators to provide required phase noise and short/long term stability.

2.3.3.1.2 Performance Specifications

See device specifications in Table 2.3.3-3.

2.3.3.1.3 Synchronizer

The purpose of the Synchronizer is to function as part of the local clock recovery in conjunction with an IEEE1588v2 software stack to provide local time of day and to generate a local synchronous reference signals to be used for synthesis of necessary clocks to other functions within the O-RU₇₋₂.

2.3.3.1.3.1 Hardware Specifications

2.3.3.1.3.1.1 Interface

Table 2.3.3-2 Synchronizer Interface Specifications

Item Name	Description
REFA, REFAA	differential or single ended reference pulse inputs.
REFB, REFBB	differential or single ended reference pulse inputs.
Status and Control	control core operation and report on state of the PLL.
XOA, XOB	complimentary reference inputs or reference crystal connections
Output0 and Output0N & Output1 and Output1N	reference outputs to feed further synthesis, transceiver or logic blocks.

2.3.3.1.3.1.2 Performance Specifications

Table 2.3.3-3 Synchronizer Performance Specifications

Item Name	Description	Notes
10 Hz Offset	−77 dBc/Hz	Device configuration: fOSC = 52 MHz crystal, fREF = 30.72 MHz, fVCO = 2457.6 MHz, fOUT = 245.76 MHz, DPLL BW = 50 Hz, internal zero delay operation
100 Hz Offset	−93 dBc/Hz	
1 kHz Offset	−114 dBc/Hz	
10 kHz Offset	−125 dBc/Hz	
100 kHz Offset	−130 dBc/Hz	
1 MHz Offset	−140 dBc/Hz	
10 MHz Offset	−156 dBc/Hz	

2.3.3.1.3.2 Hardware Design

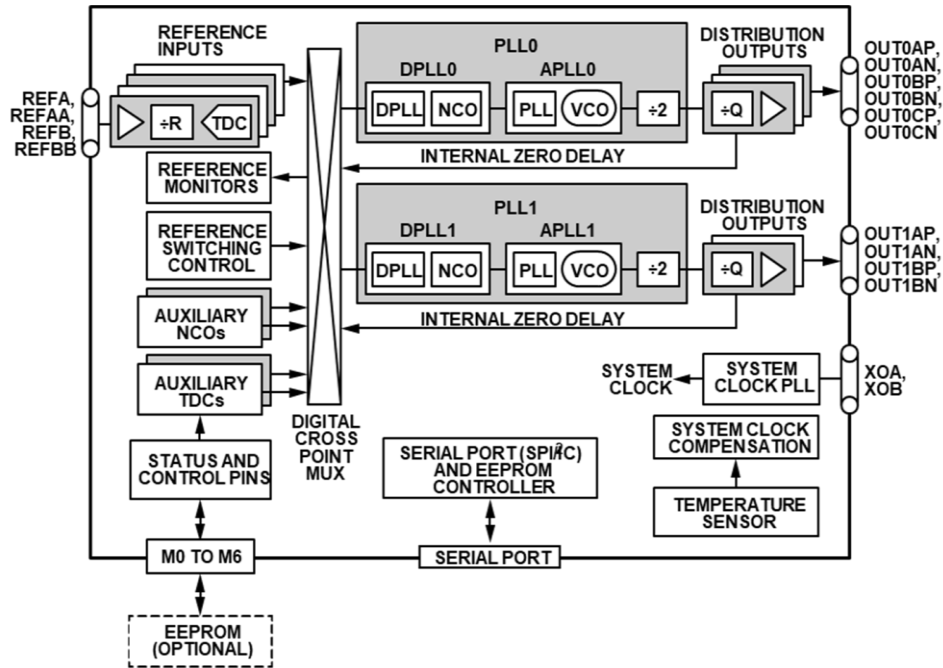


Figure 2.3.3-2 Synchronizer Block Diagram

2.3.3.1.4 Reference Synthesizer

The purpose of the clock synthesizer is to generate low jitter, high stability reference clocks as needed throughout the design.

2.3.3.1.4.1 Hardware Specifications

2.3.3.1.4.1.1 Interface

Table 2.3.3-4 Synthesizer Interface Specifications

Item Name	Description
Reference Clock(s) Input	The clock inputs device should receive a stable reference from a crystal, network reference or other suitable source. More than one input is allowed that may be selected between as required by the system.
Control	Control of the transceiver is by way of 3 or 4 wire SPI or I2C functioning as a slave. Support for 1.8V control is required and tolerance of 3.3V is preferred. Typically, the clock devices will be part of an IEEE1588v2 protocol loop controlled by way of this control interface, or other GPIO lines as required by the hardware implementation.
Clock Outputs	One or more clock outputs are required to drive the digital device and transceiver clock inputs. Each output should be independently programmable in frequency to suit the application. The outputs should nominally be differential to preserve clocking edge and to maximize tolerance of high common mode signals.

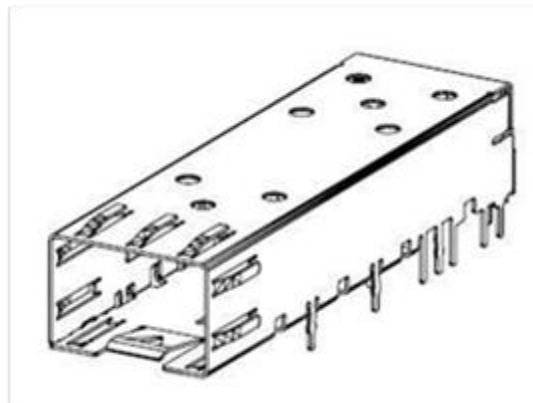
Table 2.3.4-1 External Port List

Port Name	Feature Description
Fronthaul interface	One or more 25Gbps optical interfaces including a primary link and optionally a second link for redundancy or daisy chain.
Debug interface	RJ45 for maintenance access.
Power interface	Two pin male panel mount connectors (x1)
RF Interface	4.3-10+ IP67 connector (x4) [9] [11] / M-LOC cluster/multiport RF connector [12]
Calibration Port (Optional)	N Female or 4.3-10+ Female
AISG[10] RET Port	RS-485, IEC 60130-9, or RF connectors

2.3.4.2 Hardware Design

2.3.4.2.1 Fronthaul Interface

The fronthaul interface to the OMC is provided by an optical fiber or fibers connected to the chassis by way of a SFP or SFP+ cage connector. In addition, an IP67 connector seal should be selected to provide appropriate environmental protection.


Figure 2.3.4-1 SFP+ case and connector

2.3.4.2.2 Debug Interface

The local maintenance port or debug port is by way of a standard ethernet connector. The RJ45 Ethernet connector are standardized components and available from multiple vendors. The following figure describes the general form factor of one such RJ45 connector.

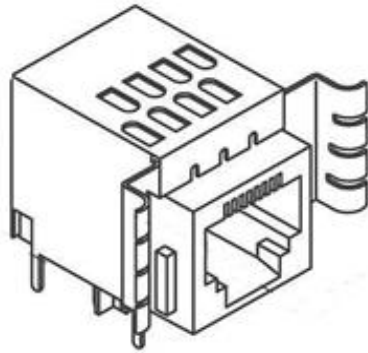


Figure 2.3.4-2 RJ45 Interface

2.3.4.2.3 Power Interface

The standard power interface is defined by a 2-pin panel mount connector as shown in the following figure.

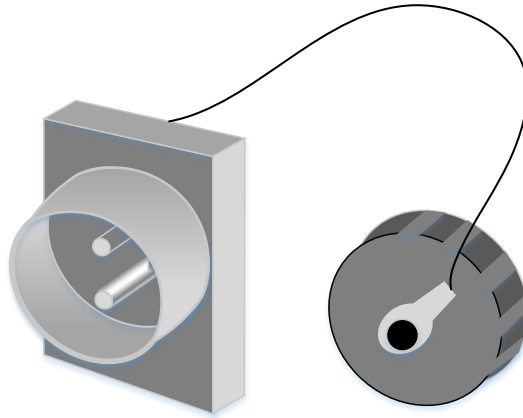
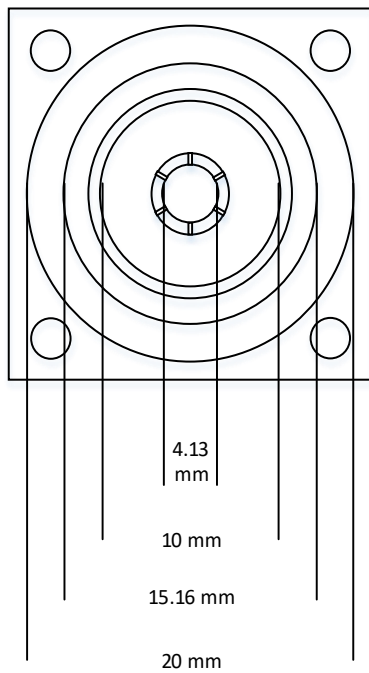


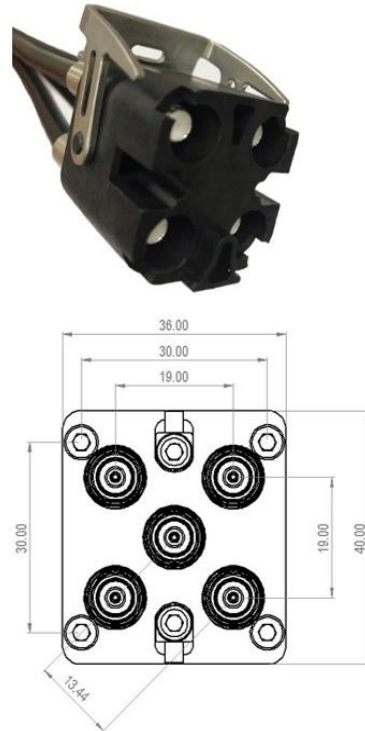
Figure 2.3.4-3 Power Interface

2.3.4.2.4 RF Interface

The standard RF interface are the 4.3-10+ Low PIM RF connector and the M-LOC cluster/multiport connector as shown in the following figure.



(a)



(b)

Figure 2.3.4-4 (a) 4.3-10+ RF Connector [9] [11] (b) M-LOC cluster/multiport connector [12]

2.3.5 Mechanical

One such mechanical structure is shown in the following figure. The shell should provide appropriate environmental protection, RF shielding and passive thermal dissipation. Internally, thermal pads are used to contact local heat sources and transfer that heat to the frame which in turn dissipates the heat through bulk metal and fins used to increase the effective surface area. Other enclosures are possible.

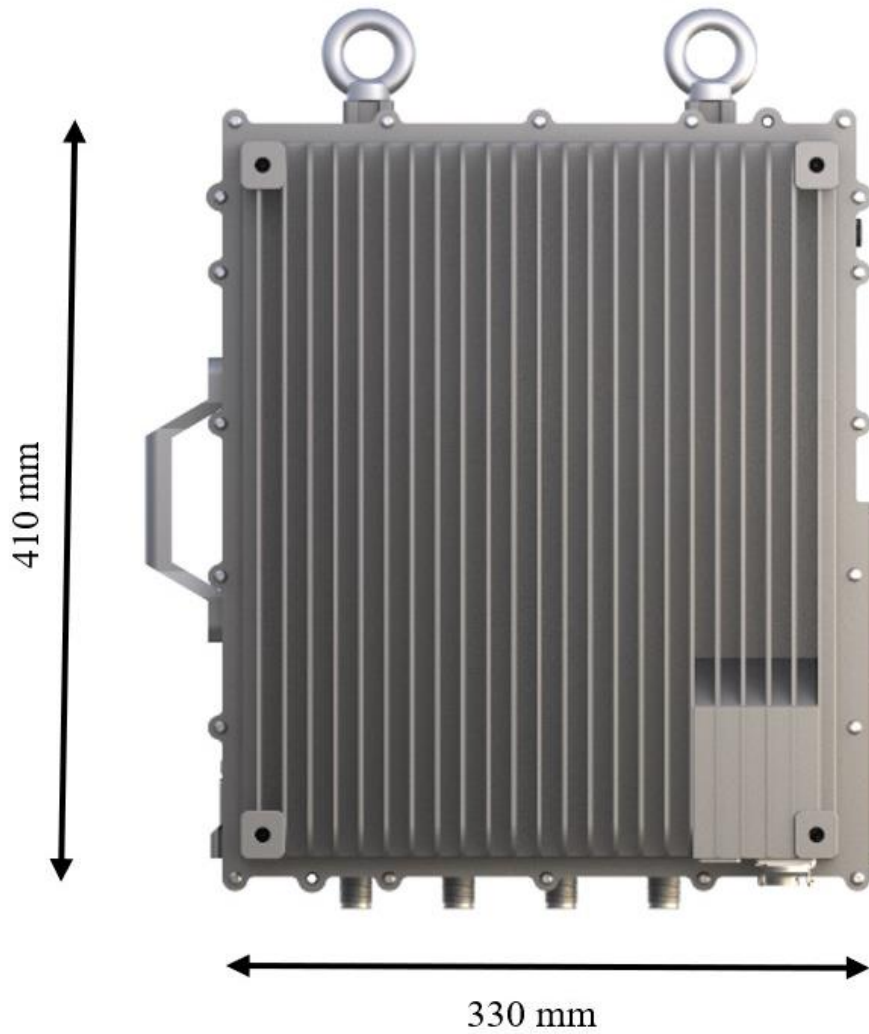


Figure 2.3.5-1 Mechanical Enclosure Side View

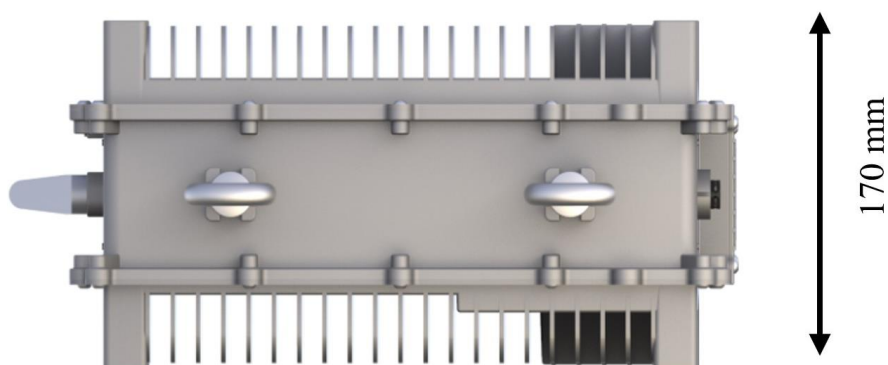


Figure 2.3.5-2 Mechanical Enclosure End View

2.3.6 Power Unit

For the Outdoor Microcell, the power solution of the O-RU₇₋₂ is provided externally from a telco -48V supply. Internally, the telco supply is converted and distributed to devices using appropriate DC-DC and LDO converters. Optionally the unit can be powered by AC line power which is not shown here.

2.3.6.1 Hardware Specifications

The power unit is responsible for isolating and converting the telco supply to appropriate voltages for distribution to individual supply domains in the O-RU₇₋₂. The typical dissipation is outlined below. The RF front end is broken out separately since different deployment scenarios will require different frequencies and power levels resulting in significantly different total dissipation. Total Power is the sum of the two sections below.

Table 2.3.6-1 Transceiver Module Specifications

Module	4T4R (Watts)		Notes
FPGA	16		
Transceiver	7		
Synchronization & Clocking	3		
DDR	1		
Others	3		
SFP+ Module	1.5		1 module installed
total device consumption	31.5		
O-RU ₇₋₂ Low PHY Transceiver power consumption	36.8		On board power efficiency 85%

Table 2.3.6-2 RF Front End Module Specifications

Module	4T4R (Watts)		Notes
PA	125		75% Tx, Max RF. 10W rms output @ 100 MHz
PA Drive Amp	4		
LNA	0.5		25% Rx
Others	2.5		Bias generators and controller
total device consumption	132		
O-RU ₇₋₂ RF Front End power consumption	155.3		On board power efficiency 85%

2.3.6.2 Hardware Design

The block diagram of the power unit is shown in the figure below, which depicts the power distribution to various functional modules such as RF PAM board and FPGA/RF SoC board (Digital) along with approximate power requirements.

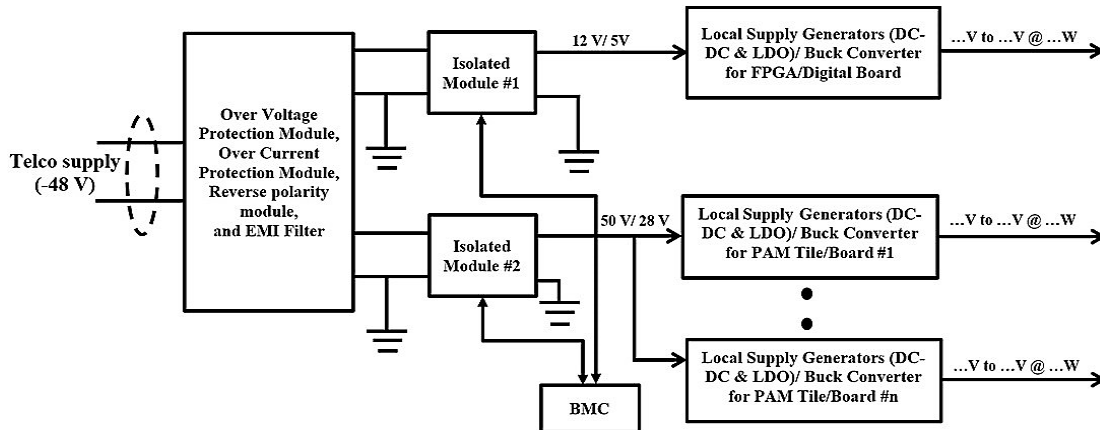


Figure 2.3.6-1 Power Reference Design

2.3.7 Thermal

Heat dissipation in O-RU7-2 is widely done through passive cooling via three modes: conduction, convection, and radiation. These mechanisms are accomplished with help of heat sink, heat pipes and vapor chamber etc.

The transfer of heat though conduction can usually be described by Fourier's law, which is stated:

$$q = (k/s)A \cdot dT$$

k is the thermal conductivity of the material, and has units of W/m·K, s = material thickness, A = heat transfer area, dT = temperature difference between inside and outside wall.

Table 2.3.7-1 Thermal Conductivity of common Metals [13]

Type of material	Thermal Conductivity (W/m.K)
Diamond	2300
Pure Al	237
Hard Aluminium 4.5% Cu	177
Au	315-317
Ag 99.9%	411-429
Iron	80
Tin	67

The heat sinks transfer heat to the space by thermal convection and radiation. For thermal convection, it can be calculated by the Newton's Equation:

$$Q_d = \alpha_c A \Delta t = \alpha_c A (t_\omega - t_f)$$

where, Q_d is the convection heat (W), t_ω is the surface temperature of object (K), t_f is the temperature of airflow (K), A is the surface area of the object involved in heat transfer (m^2), Δt is the temperature difference between the surface temperature of object and airflow (K), α_c is convective heat transfer coefficient ($W/m^2.K$).

Table 2.3.7-2 Convective heat transfer coefficient [14]

Flow Type	Heat transfer coefficients ($W/m^2.K$)
-----------	--

Forced convection; low speed flow of air over a surface	10
Forced convection; moderate speed flow of air over a surface	100
Forced convection; moderate speed cross flow of air over a cylinder	200
Forced convection; moderate flow of water in a pipe	3000

For thermal radiation, it can be calculated by the following Equation:

$$Q_r = \varepsilon \delta A (t_w^4 - t_f^4)$$

where, Q_r is the radiation heat (W), ε is the thermal emissivity, δ is the Stefan–Boltzmann constant, its value is $5.67032 \times 10^{-8} \text{ W/ (m}^2 \cdot \text{K}^4\text{)}$. From these Equations, we can see that the increase of surface area A and thermal emissivity ε will improve the heat dissipation.

Different materials can have widely different emissivity values within the range of 0 to 1.00.

Material	Emissivity Value (ε)
Aluminium: anodized	0.77
Aluminium: polished	0.05
Brass: highly polished	0.03
Brass: oxidized	0.61
Carbon: candle soot	0.95
Carbon: graphite, filed surface	0.98
Copper: polished	0.05
Copper: oxidized	0.65

2.3.8 Environmental and Regulations

The O-RU₇₋₂ hardware system is RoHS Compliant. Unit is passively cooled.

Table 2.3.8-1 Environmental Features

Item Name	Description	Notes
Operating Temperature	-40°C to +55°C	Cold and hot start
Operating Relative Humidity	5% to 95%	
Atmospheric Pressure	70 to 106Kpa	

2.4 Integrated O-DU & O-RU (gNB-DU) Reference Design

2.4.1 O-DU Portion of Integrated Reference Design

2.4.1.1 O-DU High-Level Functional Block Diagram

2.4.1.2 O-DU Hardware Components

2.4.1.2.1 Digital Processing Unit

2.4.1.2.2 Hardware Accelerator (if required by design)

2.4.1.2.2.1 Accelerator Design 1

2.4.1.2.2.2 Accelerator Design 2

2.4.1.3 Synchronization and Timing

2.4.2 O-RU Portion of Integrated Reference Design

2.4.2.1 O-RU High-Level Functional Block Diagram

2.4.2.2 O-RU Hardware Components

2.4.2.2.1 Digital Processing Unit

2.4.2.2.2 RF Processing Unit

2.4.2.2.2.1 Transceiver Reference Design

2.4.2.2.2.2 Power Amplifier (PA) Reference Design

2.4.2.2.2.3 Low Noise Amplifier (LNA) Reference Design

2.4.2.2.2.4 RF Switch Reference Design

2.4.2.2.2.5 Antenna / Phased Array Reference Design

2.4.2.3 Synchronization and Timing

2.4.3 External Interface Ports

2.4.4 O-DU/O-RU Firmware (if required by design)

2.4.5 Mechanical

2.4.6 Power Unit

1 2.4.7 Thermal

2 2.4.8 Environmental and Regulations

3 2.5 Integrated O-CU & O-DU Reference Design

4 2.5.1 O-CU Portion of Integrated Reference Design

5 2.5.1.1 Integrated O-CU High-Level Functional Block Diagram

6 2.5.1.2 O-CU Hardware Components

7 2.5.1.2.1 Digital Processing Unit

8 2.5.1.2.2 Hardware Accelerator (if required by design)

9 2.5.1.2.2.1 Accelerator Design 1

10 2.5.1.2.2.2 Accelerator Design 2

11 2.5.1.3 Synchronization and Timing

12 2.5.1.4 O-CU Firmware (if required by design)

13 2.5.2 O-DU Portion of Hardware Reference Design

14 2.5.2.1 O-DU High-Level Functional Block Diagram

15 2.5.2.2 O-DU Hardware Components

16 2.5.2.2.1 Digital Processing Unit

17 2.5.2.2.2 Hardware Accelerator (if required by design)

18 2.5.2.2.2.1 Accelerator Design 1

19 2.5.2.2.2.2 Accelerator Design 2

20 2.5.2.3 Synchronization and Timing

21 2.5.3 External Interface Ports

22 2.5.4 O-DU Firmware

23 2.5.5 Mechanical

2.5.6 Power Unit

2.5.7 Thermal

2.5.8 Environmental and Regulations

2.6 FHGW Hardware Reference Design

2.6.1 FHGW High-Level Functional Block Diagram

2.6.2 FHGW Hardware Components

2.6.2.1 Digital Processing Unit

2.6.3 Synchronization and Timing

2.6.4 External Interface Ports

2.6.5 Mechanical

2.6.6 Power

2.6.7 Thermal

2.6.8 Environmental

Annex 1: Parts Reference List

Annex 1.1 O-CU Reference Parts List

The tables below represent potential options for component selection for O-CU₇₋₂.

The HW reference design of O-CU is identical to the O-DU₇₋₂ except for the need of HW accelerator as outlined in section 2.2.

Annex 1.2 O-DU₇₋₂ Reference Parts List

The tables below represent potential options for component selection for O-DU₇₋₂.

Intel Based O-DU₇₋₂

No.	Item Name	Description
1	Intel® Xeon® Processor D-2183IT	Intel® Xeon® Skylake D D-2100 SoC processor
2	Intel® I350	Intel® 1GbE x4 Ethernet controller
3	Intel® X557-AT2	Intel® 10GbE x2 Ethernet controller
4	Intel® I210	Intel® 1GbE Ethernet controller
5	ASPEED® AST-2500	ASPEED® BMC controller
6	UniIC® SCQ08GU03H1F1C-26V	UniIC® DDR4 8GB
7	Powerleader® MTFDDAK480TDC-1AT1ZABYY	Powerleader® SSD 480GB
8	Accelink® RTXM228-551	Accelink® 10Gbps Optical module

Annex 1.3 O-RU₇₋₂ Reference Parts List

The tables below represent potential options for component selection for O-RU₇₋₂.

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ADI Based O-RU₇₋₂

No.	Item Name	Description
1	ADP196ACBZ-R7	5V, 3A Logic Controlled High-Side Power Switch
2	LTC7150SEY#PBF	20V, 20A Synchronous Ste-Down Regulator
3	ADRV9029	Integrated, Quad RF Transceiver with Observation Path and DPD/CFR
4	ADP223ACPZ-R7	Dual, 300 mA Adjustable Output, Low Noise, High PSRR Voltage Regulator
5	LTC4217IDHC#PBF	2A Integrated Hot Swap Controller
6	ADCLK846BCPZ-REEL7	1.8V, 6 LVDS/12 CMOS Output Low Power Clock Fanout Buffer
7	AD9545BCPZ	IEEE 1588 V2 and 1 pps Synchronizer and Adaptive Clock Translator
8	ADRF5545A	Dual-Channel 2.4 to 4.2 GHz Receiver Front End
9	LTC3315AIV#TRMPBF	Dual 5V, 2A Synchronous Step-Down DC/DCs
10	LTC3634EUFD#PBF	15V Dual 3A Monolithic Step-Down Regulator for DDR Power
11	LTC2928IUHF#BPF	Multichannel Power Supply Sequencer and Supervisor
12	ADP5054ACPZ-R7	Quad Buck Regulator Integrated Power Solution
13	AD9576BPZ	Dual PLL, Asynchronous Clock Generator
14	LT1389ACS8-1.25#PBF	Nanopower Precision Shunt Voltage Reference
15	Intel® Arria® 10 SoC	Intel® Arria® 10 SoC 1AS048F35
16	MAX1619MEE	Temp Sensor with Dual-Alarm Outputs and PMBus

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